
UNIT 1: INTRODUCTION TO MANAGEMENT INFORMATION SYSTEM (MIS)

★ Introduction

A number of trends affect today's business environment. One of them is the transition from an industrial economy to an information services economy. Beginning with the industrial revolution, productivity gains in the westerns economy were tied to industrial production and manufacturing of goods and services. Since the 1960s, 50 percent of all productivity gains have been attributable to the use of information technology. The ability to capture, store, process, and distribute information is critical to most organization.

Today, the traditional organization is being transformed in to the information based organization, which uses information and information technology to produce significant changes in work patterns. The organization of the past was highly structured and composed of many different functions. In this type of organization, each unit maintained its own information. The organization of the future will have a flexible, changeable structure. Teams consisting of specialists from various functional areas will work together on projects that address new market opportunities. Shared information data bases will link individuals to each other. In the future, manager in the information based organization will become responsible for using and managing technology. They will use information technology (IT) as a tool to provide effective customer service, analyze market opportunities, and manage production and manufacturing operations. IT will become an integral part of business. To understand the information based company of the future, let's first learn why traditional organizations are evolving into information based organizations.

Unit Objectives: After studying this unit, you should be able to:

- Explain the development of information system
- Define management, information, system and finally management information system
- Understand the impact of information technology
- Explain the purpose of information technology
- Describe the features of information

1.1. Definition and Importance of MIS

Management: Management has been defined in a variety of ways, but for our purposes it comprises the process of activities that describe what managers do in the operation of their organization: plan, organize, initiate, and control operations. They plan by setting strategies and goals and selecting the best course of action to achieve the plan. They organize the tasks necessary for the operational plan, set these tasks up into homogeneous groups, and assign authority delegation. They control the performance of the work by setting performance standards and avoiding deviations from standard.

Because **decision making** is such a fundamental prerequisite to each of the foregoing processes, the job of an MIS becomes that of facilitating decisions necessary for planning, organizing, and controlling the work and functions of the business.

Information: Data must be distinguished from information, and this distinction is clear and important for our purpose. Data are facts and figures that are not currently being used in a decision process and usually take the form of historical records that are recorded and filed without immediate intent to retrieve for decision-making.

Data are collections of facts or events represented in the form of symbols, such as digits, alphabets, pictures, graphs, etc. Capturing, processing and storage of data are the essential functions of any IT infrastructure. Data are the basic raw materials in the process of generation of information. Data may be collected from internal sources as well as external sources. Information consists of data that have been retrieved, processed, or otherwise used for informative or inference purposes, argument, or as a basis for forecasting or decision-making.

Information is knowledge that one derives from facts placed in the right context with the purpose of reducing uncertainty. From the manager's point of view, information serves the purpose of reducing uncertainty regarding the alternative course of action, in the process of decision-making. Availability of information regarding the alternative improves the odds in favor of making a correct decision. Information is recognized as one of the most important corporate resources.

1.2. Subsystems of MIS

MIS has been introduced as a broad concept referring to a federation of subsystems. Two approaches to defining the subsystems of an MIS are according to the organizational functions, which they support, and according to managerial activities for which they are used.

Organizational Function Subsystems

Because organizational functions are somewhat separable in terms of activities and are defined managerially as separate responsibilities, MIS may be viewed as a federation of information system – one for each major organizational function. There may be common support systems used by more than one subsystem, but each functional system is unique in its procedures, programs, models, etc., typical subsystems for a business organization engaged in manufacturing are:

Major functional subsystem	Some typical uses
Marketing	Sales forecasting, sales planning, customer and sales analysis
Manufacturing	Production planning and scheduling, cost control analysis
Logistics	Planning and control of purchasing, inventories, distribution
Personnel	Planning personnel requirements, analyzing performance, salary administration
Finance & accounting	Financial analysis, cost analysis, capital requirements planning income measurement
Information processing	Information system planning, cost effectiveness analysis
Top management	Strategic planning, resource allocation

The database is the primary means of integration of the various subsystems. A data item that is stored or updated by one subsystem is then available to the other subsystems. For instance, the sales and inventory information used by the marketing subsystem is supplied through the logistical subsystem; the same data is used by the manufacturing subsystem for production planning and scheduling.

Activities Subsystem

Another approach to understanding the structure of an information system is in terms of the subsystem, which perform various activities. Some of the activities subsystems will be useful for more than one organizational function subsystem; others will be useful for only one function.

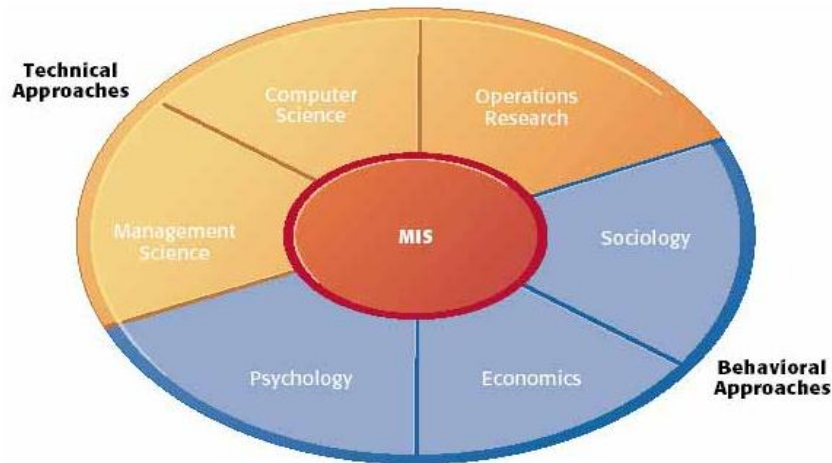
Activity subsystem	Some typical uses
Transaction processing	Processing of orders, shipments, and receipts
Operational control	Scheduling of activities and performance reports
Management control	Formulation of budgets and resource allocation
Strategic planning	Formulation of objectives and strategic plans

Inputs, processes and output of MIS at all levels of Management

	Inputs	Process	Outputs
Strategic	Plans, competitor information, overall market information	Summarize Investigate Compare Forecast	Key ratios, ad hoc market analysis, Strategic plans
Tactical	Historical, and budget data	Compare Classify Summarize	Variance analysis Exception report
Operational	Customer orders Programmed stock Control levels Cash receipt or payments	Update files Output reports	Updated files Listings, invoices

1.3. Contemporary Approaches to MIS

Multiple perspectives on information systems show that the study of information systems is a multidisciplinary field. No single theory or perspective dominates. The following figure illustrates the major disciplines that contribute problems, issues, and solutions in the study of information systems. In general, the field can be divided into technical and behavioral approaches. Information systems are socio-technical systems. Though they are composed of machines, devices, and “hard” physical technology, they require substantial social, organizational, and intellectual investments to make them work properly.



Technical Approach

The technical approach to information systems emphasizes mathematically based models to study information systems, as well as the physical technology and formal capabilities of these systems. The disciplines that contribute to the technical approach are computer science, management science, and operations research.

Computer science is concerned with establishing theories of computability, methods of computation, and methods of efficient data storage and access. Management science emphasizes the development of models for decision-making and management practices. Operations research focuses on mathematical techniques for optimizing selected parameters of organizations, such as transportation, inventory control, and transaction costs.

Behavioral Approach

An important part of the information systems field is concerned with behavioral issues that arise in the development and long-term maintenance of information systems. Issues such as strategic business integration, design, implementation, utilization, and management cannot be explored usefully with the models used in the technical approach. Other behavioral disciplines contribute important concepts and methods.

For instance, sociologists study information systems with an eye toward how groups and organizations shape the development of systems and also how systems affect individuals, groups, and organizations. Psychologists study information systems with an interest in how human decision makers perceive and use formal information. Economists study information systems with an interest in what impact systems have on control and cost structures within the firm and within markets.

The behavioral approach does not ignore technology. Indeed, information systems technology is often the stimulus for a behavioral problem or issue. But the focus of this approach is generally not on technical solutions. Instead, it concentrates on changes in attitudes, management and organizational policy, and behavior.

Throughout this book you will find a rich story with four main actors: suppliers of hardware and software (the technologists); business firms making investments and seeking to obtain value from the technology; managers and employees seeking to achieve business value (and other goals); and the contemporary legal, social, and cultural context (the firm's environment). Together these actors produce what we call management information systems.

The study of management information systems (MIS) arose in the 1970s to focus on the use of computer-based information systems in business firms and government agencies (Laudon, 1974; Davis and Olson, 1985). MIS combines the work of computer science, management science, and operations research with a practical orientation toward developing system solutions to real-world problems and managing information technology resources. It is also concerned with behavioral issues surrounding the development, use, and impact of information systems, which are typically discussed in the fields of sociology, economics, and psychology. The study of information systems has just started to influence other disciplines (Baskerville and Myers, 2002) through concepts such as the information processing view of the firm.

Our experience as academics and practitioners leads us to believe that no single approach effectively captures the reality of information systems. The successes and failures of information systems are rarely all technical or all behavioral. Our best advice to students is to understand the perspectives of many disciplines. Indeed, the challenge and excitement of the information systems field is that it requires an appreciation and tolerance of many different approaches.

In the socio-technical view of systems, optimal organizational performance is achieved by jointly optimizing both the social and technical systems used in production (Mumford, 2000 and 1997; Williams and Edge, 1996).

Adopting a socio-technical systems perspective helps to avoid a purely technological approach to information systems. For instance, the fact that information technology is

rapidly declining in cost and growing in power does not necessarily or easily translate into productivity enhancement or bottom-line profits. The fact that a firm has recently installed an enterprise-wide financial reporting system does not necessarily mean that it will be used, or used effectively. Likewise, the fact that a firm has recently introduced new business procedures and processes does not necessarily mean employees will be more productive in the absence of investments in new information systems to enable those processes.

1.4. The Evolution and Characteristics of the Information Age

Historically, organizations have become hierarchies of complex functions over time. One of the fundamental principles influencing the evolution of industrial organizations was specialization of labor. The division of labor led to the fragmentation of work, with workers in many different areas – marketing, manufacturing, accounting, and so on – performing specialized tasks.

A second factor in the development of modern organizations is the command-and-control structure that can be traced back to early railroad companies, which required predictable, safe control systems. Today's business bureaucracy, with its formalized operating procedures and formal lines of authority, evolved from these early railroad organizations. The standard pyramid, with work broken down by departments – each with its own budget and control systems – is still a common organizational form today.

The third factor influencing the performance of modern organizations was the nature of the post-World War II market: It was a seller's market. Given the unrelenting demand for goods and services, customers were willing to buy anything that was available. Customer service was not necessarily a critical success factor.

In the 1990s, the nature of the market has changed. Customers have power over suppliers that they didn't have before. Part of this customer power comes from access to information. When prospective car buyers can look up dealer invoice cost on a new auto in Consumer Reports, they gain new leverage over the dealer.

Information technology provides a means to focus on the customer. By carefully tracing and analyzing customer buying behavior, companies like Wal-Mart can allocate inventory to meet customer needs. They can also use information technology to provide better customer service. Some companies have a customer service network that route

customer calls to the same service representative again and again, thus creating a sense of personal service.

Other companies use a service management network to provide on-line technical service for large number of services. When a support call is dispatched to service technicians, the system automatically provides data on the maintenance history of a particular service area. Many microcomputer vendors give their customers access to bulletin board systems that provide up-to-date technical information on product features.

Finally, information technology is transforming the scope of doing business. Worldwide communications networks enable businesses to operate in global markets and to reach new customers. Information technology also makes it possible to do business 24 a day. Individuals on project teams may work in distant locations and interact using electronic communication networks and videoconferencing facilities.

1.5. What Is Information Technology?

In the 1960s and 1970s, the term information technology (IT) was a little known phrase that was used by those who worked in places like banks and hospitals to describe the processes they used to store information. With the paradigm shift to computing technology and "paperless" workplaces, information technology has come to be a household phrase. It defines an industry that uses computers, networking, software programming, and other equipment and processes to store, process, retrieve, transmit, and protect information.

In the early days of computer development, there was no such thing as a college degree in IT. Software development and computer programming were best left to the computer scientists and mathematical engineers, due to their complicated nature. As time passed and technology advanced, such as with the advent of the personal computer in the 1980s and its everyday use in the home and the workplace, the world moved into the information age.

By the early 21st century, nearly every child in the Western world, and many in other parts of the world, knew how to use a personal computer. Businesses' information technology departments have gone from using storage tapes created by a single computer operator to interconnected networks of employee workstations that store information in a

server farm, often somewhere away from the main business site. Communication has advanced, from physical postal mail, to telephone fax transmissions, to nearly instantaneous digital communication through electronic mail (email).

Great technological advances have been made since the days when computers were huge pieces of equipment that were stored in big, air conditioned rooms, getting their information from punch cards. The information technology industry has turned out to be a huge employer of people worldwide, as the focus shifts in some nations from manufacturing to service industries. It is a field where the barrier to entry is generally much lower than that of manufacturing, for example. In the current business environment, being proficient in computers is often a necessity for those who want to compete in the workplace.

Jobs in information technology are widely varied, although many do require some level of higher education. Positions as diverse as software designer, network engineer, and database administrator are all usually considered IT jobs. Nearly any position that involves the intersection of computers and information may be considered part of this field.

In today's up-and-down job market and shaky economy, there are few career fields that are as consistent as the field of Information Technology. As a branch of engineering, Information Technology specializes specifically in using computer technology and telecommunications technology to transmit, translate and store various types of information- although there are a ton of other types of jobs that do fall under the Information Technology umbrella.

Essentially information technology is all about using information as a commodity or resource, just like any other job. The information that is being translated, disseminated, stored, transmitted, etc. is usually in the forms of audio, video, textual and numerical information and is processed through the use of microelectronics and computers.

The field of Information Technology is usually used to describe a whole series of jobs, but in reality, there are tons of jobs that are called Information Technology jobs, but are actual part of a subcategory like Management Information Services and Information Services.

Once upon a time, the Information Technology center of a business would have been composed of just one guy sitting at a computer all day. But, as computer technology has advanced, and information has become more valuable, the Information Technology center grew with it. Now, a company will either have a gigantic IT center, or they may need so many specialists that they actually contract out the Info-tech needs to an entirely separate company. Now, in order to successfully run an Information Technology center, a company would need at least a database management system, a cryptographer, some system administrators, some database administrators, at least one information manager and a Chief Information Office (CIO,) who is the head honcho of the whole shabang.

1.6. The Career Side of Information Technology

Each of the management levels – operational, tactical and strategic planning requires different information systems with the characteristics discussed here below:

1. Operational Systems

At the operational level the primary concern is to collect, validate, and record transactional data describing the acquisition or disbursement of corporate resources. Financial data on accounts receivable, accounts payable, payroll, and cash receipts need to be recorded as they occur. When a sale is transacted, data on the items ordered are recorded, the inventory level for these items is adjusted, a shipping label and packing slip are prepared and an invoice is generated. The original transaction – the sale of the item – creates numerous transactions in order processing, inventory, and billing. Operational level information systems often have the following characteristics.

- ✓ **Repetitiveness:** the information they produce is usually generated repetitively at periodic intervals, such as daily, weekly, or monthly.
- ✓ **Predictability:** the information they produce usually does not contain any surprises or unexpected results for the manager or other users of the system. That is, people are paid what they were expected to be paid and customers are billed for what they purchased during the month.
- ✓ **Emphasis on the past:** the information produced usually describes past activities of the organization. For example, the output of a payroll system describes employees' work. The checks to vendors describe past purchases by the organization. Customer invoices describe past sales to customers. Stock reports describe past changes in inventory.
- ✓ **Detailed nature:** the information produced is very detailed. That is, paychecks provide detailed information on the workweek of each employee and the specifics

-
- of each employee's gross and net pay. Customer invoices specify details regarding purchases made during the period, the terms under which the purchases must be paid, and the total amount, taxes, and other charges due.
- ✓ **Internal origin:** the data for operational systems usually spring entirely from internal sources. That is, the data for paychecks come from internal documents such as time cards and employee master records. The data for customer invoices come from sales orders and shipping documents.
 - ✓ **Structured form:** the form of the data used as input and the form of the information produced by operational level systems usually very structured. That is, the data on time cards are carefully formatted in identical fashion on each time card. Or the data on each customer invoice are carefully formatted in identical fashion. In short, the form and the format of the data input and the information output of the systems are highly structured.
 - ✓ **Great accuracy:** the accuracy of the data used as input to such systems is usually very high. The data input and information output are carefully checked in a variety of ways.

2. Tactical Systems

The second level in the framework consists of tactical systems. Tactical systems provide middle-level managers with the information they need to monitor and control operations and to allocate their resources more effectively. In tactical systems, transaction data are summarized, aggregated, or analyzed. Tactical systems are designed to generate a variety of reports, including summary reports, exception reports, and ad hoc reports.

1. **Summary reports:** it provide management with important totals, average, key data, and abstracts on the activities of the organization (e.g., List Total regular and overtime hours earned at each plant, total weekly sales, by salesperson, by product, and sales region).
2. **Exception reports:** it warns managers when results from a particular operation have exceeded or not met the expected standard for the organization (sales fall by 10% for some employees, more over time in certain plants).
3. **Ad hoc reports** that managers need, usually quickly, that may never be needed again. Ad hoc reports present information that the manager needs to solve a unique problem.

Tactical information systems differ from operational information systems in their basic purpose. Their purpose is not to support the execution of operational tasks but to help the

manager control these operations. Tactical information systems often have the following features.

1. **Periodic nature:** the information from a tactical system is sometimes produced periodically (e.g., accounts receivable report). Sometimes ad hoc information may also be produced.
2. **Unexpected findings:** the information produced by a tactical information system may not be the information that was expected to be produced.
3. **Comparative nature:** the information produced is usually comparative in nature rather than merely descriptive. Tactical information systems should provide managers with information that alerts them to variances from accepted standards or results that are not within the normal range, so that remedial actions can be taken swiftly.
4. **Summary form:** the information produced is usually not detailed but in summary form. The credit manager is not interested in a detailed listing of each customer account and its balance.
5. **Both internal and external sources:** the data used for input to the system may not be confined to source internal to the organization.

3. Strategic Information Systems

Strategic planning level information systems are designed to provide top management with information that assists them in making long range planning decisions for the organization; Strategic planning information systems often have these characteristics:

1. **Ad Hoc Basis:** the information may be produced either regularly or periodically. For example, periodic accounting system reports are used by top management in its planning function. However, strategic planning information is more often produced when it is needed, on an ad hoc basis.
2. **Unexpected Information:** the information produced by the system may not be the information that was anticipated.
3. **Predictive Nature:** the information produced is usually predictive of future events rather than descriptive of past events. Long-range planners try to set a course for an organization through an uncharted future. Their primary task is to choose a route that will improve the organization's level of success. The information that the strategic planning system provides should help these planners the risks involved in their choice of routes.

4. **Summary Form:** the information produced is usually not detailed but in summary form. Long range planners are not usually interested in detailed information. They are usually concerned with more global data. (Example, buying trends, overall demographic characteristics of groups of customers).
5. **External Data:** a large part of the data used for input to the system may be acquired from sources external to the organization. (Investment opportunities, rates of borrowed capital, demographic characteristics of a market make group and economic conditions must be obtained from data maintained outside the organization).
6. **Unstructured Format:** the data used for input to the system may contain data that are unstructured in format. For instance, forecasts of future market trends may use the opinions of store buyers, salespeople, or market analysts obtained in causal conversations.
7. **Subjectivity:** the data used for input the system may be highly subjective and their accuracy may be suspect. For instance, forecasts of future stock market trends may be based partly on rumors reported by brokers. Forecasts of the expected market share of your organization within the industry observers who are basic their opinion on rumors and on conversations held with a variety of industry personnel are examples of subjective information.

A comparison of Information Systems at the operational, tactical and strategic planning levels.

Characteristic	Summary of Classification Information Systems		
	Operational	Tactical	Strategic Planning
Frequency	Regular, repetitive	Most regular	Often ad hoc
Dependability of results	Expected results	Some surprises may occur	Results often contain surprises
Time period covered	The past	Comparative	Predictive of the future
Level of detail	Very detailed	Summaries of data	Summaries of data
Source of data	Internal	Internal and external	Mostly external
Nature of data	Highly structured	Some of unstructured data	Highly unstructured
Accuracy	Highly accurate data	Some subjective data	Highly subjective data
Typical user	First line supervisors	Middle managers	To management
Level of decision	Task oriented	Control and resource allocation oriented	Goal oriented

? Review Question

1. What is Management information System?
2. What is the purpose of information technology and explain its impact from individual, organizational and functional unit view?
3. Discuss the features of information as a corporate resource?
4. Explain the process of generation of information.
5. Identify and explain the two forms of software.

Summary

Management has been defined in a variety of ways, but for our purposes it comprises the process of activities that describe what managers do in the operation of their organization: plan, organize, initiate, and control operations.

Because **decision making** is such a fundamental prerequisite to each of the foregoing processes, the job of an MIS becomes that of facilitating decisions necessary for planning, organizing, and controlling the work and functions of the business.

MIS has been introduced as a broad concept referring to a federation of subsystems. Two approaches to defining the subsystems of an MIS are according to the organizational functions, which they support, and according to managerial activities for which they are used.

Part I: True/False Questions

Read the following sentences carefully and write TRUE if the statement is correct or FALSE if the statement is incorrect in your notebook.

1. Tactical systems provide middle-level managers with the information they need to monitor and control operations and to allocate their resources more effectively.
2. Data are collections of facts or events represented in the form of symbols, such as digits, alphabets, pictures or graphs
3. The information which is produced always should be descriptive in its nature.
4. Management Information System has been defined as federation of subsystems.

Part II: Multiple Choice Questions

Choose the correct answer for the following questions and write the capital letter of your correct choice in your notebook.

- Which one of the following report is included under Tactical system?
 - Summary reports
 - Exception reports
 - Ad hoc reports
 - All
 - None
- Which one of the following is the use of Transaction processing subsystem?
 - Processing of orders, shipments, and receipts
 - Formulation of objectives and strategic plans
 - Scheduling of activities and performance reports
 - Formulation of budgets and resource allocation
 - None
- Which one of the following is the use of Operational control subsystem?
 - Processing of orders, shipments, and receipts
 - Formulation of objectives and strategic plans
 - Scheduling of activities and performance reports
 - Formulation of budgets and resource allocation
 - None
- Which one of the following report presents information that the manager needs to solve a unique problem?
 - Summary reports
 - Exception reports
 - Ad hoc reports
 - All
 - None

🔑 Answer Key to Self Test Exercise

Part I: True/False Question

1. True
2. True
3. False
4. True

Part II: Multiple Choice Questions

1. D
2. A
3. C
4. C

UNIT 2: FOUNDATIONAL CONCEPTS IN MIS

★ Introduction

A management information system (MIS) provides information that organizations require to manage themselves efficiently and effectively. Management systems are typically computer systems used for managing five primary components: 1.) *Hardware*, 2.) *Software*, 3.) *Data (information for decision making)*, 4.) *Procedures (design, development and documentation)*, and 5.) *People (individuals, groups, or organizations)*. Management information systems are distinct from other information systems, in that they are used to analyze and facilitate strategic and operational activities. Academically, the term is commonly used to refer to the study of how individuals, groups, and organizations evaluate, design, implement, manage, and utilize systems to generate information to improve efficiency and effectiveness of decision making, including systems termed decision support systems, expert systems, and executive information systems.

Unit Objectives: After studying this unit, you should be able to:

- Acquainted with the foundational concepts in MIS
- Understand business management functions
- Identify the difference and similarity between data, information, knowledge and wisdom
- Recognize the information needs and sources of information for managers
- Understand the framework for Information Systems

2.1. Introduction

The MIS is an idea which is associated with man, machine, marketing and methods for collecting information's from the internal and external source and processing this information for the purpose of facilitating the process of decision-making of the business. MIS is not new, only the computerization is new, before computers MIS techniques existed to supply managers with the information that would permit them to plan and control business operations. The computer has added on more dimensions such as speed, accuracy and increased volume of data that permit the consideration of more alternatives in decision-making process.

The scope and purpose of MIS is better understood if each part of them is defined individually, thus

1. Management: Management has been define in process or activities that describe what managers do in the operation for their organization plan, organize, initiate and control operations. They plan by setting strategies and goals and selecting the best course of action to achieve the goals. They organize the necessary tasks for the operational plan, set these tasks up into homogenous groups and assign authority delegation; they control the performance standards and avoiding deviation from standard.

The decision-making is a fundamental prerequisite of each of the foregoing process, the job of MIS is facilitating decisions necessary for planning, organizing and controlling the work and functions of the business so that specified goals of business are achieved.

2. Information: Data must be distinguished from information and the distinction is clear and important for present purpose. Data are facts and figures that are not currently being used in a decision-making process and usually are taken from the historical records that are recorded and filled without immediate intent to retrieve for decision-making.

Information consists of data that have been retrieved, processed or otherwise used for information or interference purpose, argument or as a basis forecasting or decision-making regarding any business unit. Information is knowledge that one derives from facts for effective functioning of systems placed in the right context with the purpose of reducing uncertainty regarding the alternative courses of action as they are based on description and measurement of attributes of various entities associated with the enterprise.

3. System: The system can be described as a set of elements joined together for a common objective. A subsystem is a part of a larger system with which one is concerned. All systems for our purpose the organization is the system and the parts (divisions, departments, functions, unit etc) are the subsystem.

The system concept of MIS is, therefore one of optimizing the output of the organization by connecting the operating subsystems through the medium of information exchange.

Information systems and technologies are a vital component of successful businesses and organizations. It is just as important to have a basic understanding of information systems as it is to understand any other functional area in business.

Information technologies, including Internet-based information systems, are playing a vital and expanding role in business. Information technology can help all kinds of businesses improve the efficiency and effectiveness of their business processes, managerial decision making, and workgroup collaboration, thus strengthening their competitive positions in a rapidly changing marketplace. This is true whether information technology is used to support product development teams, customer support processes, electronic commerce transactions, or any other business activity. Internet-based information technologies and systems have become a necessary ingredient for business success in today's dynamic global environment.

Importance of information systems (IS) and information technology (IT). Virtually no business or organization, large or small, can compete without information systems and information technology. In its simplest form, a system is a set of interrelated components, with a clearly defined boundary, working together to achieve a common set of objectives.

An information system (IS) can be any organized combination of people, hardware, software, communications networks, and data resources that stores and retrieves, transforms, and disseminates information in an organization. People have relied on information systems to communicate with each other using a variety of physical devices (hardware), information processing instructions and procedures (software), communications channels (networks), and stored data.

2.2. Business and Management Functions

Management Information Systems (MIS) not only include software systems, but the entire set of business processes and resources that are used to pull together information from functional or tactical systems. Data is then presented in a user-friendly and timely manner so that mid and upper-level managers can use it to take the right actions. The entire system is designed so that the company will meet its strategic and tactical goals.

Significance

Organizations have multiple functional systems. These usually include sales systems, call center systems, financial systems, inventory systems, logistic systems and more. MIS combines information from multiple systems. This helps management staffers better understand their own departments' contributions. In many cases, the combination of data, such as sales figures combined with available inventory, help the manager take the appropriate action in order to meet the customer's needs.

Function

The primary function of MIS is to help a manager take an action, answer a question or ask the right question. The questions or actions should directly relate to tactical or strategic goals. A sales manager who uses projections from the financial systems to compare with actual sales from the sales system can better gauge whether goals will be met. If the target is not going to be met, then the manager and his group can review their past actions and make necessary changes in order to increase sales and meet goals.

Types

MIS is not necessarily a specific combination of functional systems, but instead is created based upon the business need. CRM (Customer Relationship Management) systems combine data that relates directly to the customer experience. ERP (Enterprise Resource Systems) combine data used in the entire sales process. Decision Support Systems or Data Warehouse often combine summary data from multiple systems in order to show executives a snapshot view of the entire organization.

Considerations

Prior to starting an MIS project, organizations need to carefully review the transactional systems, the business processes and the needs of management within an organization. As an MIS project grows, so does the cost of implementing a solution, managing the information processes and monitoring daily activities. The result of an MIS project must provide value back to the organization in order to be worth the cost.

Misconceptions

Many managers mistakenly believe that, for MIS to be effective, all data from all systems must be combined. The value of MIS is based upon how much it can help managers

manage. If this means bringing just the data needed from several systems and ignoring the rest for now, the end result still has worth, which is the ultimate goal of MIS.

The role of the MIS in an organization can be compared to the role of heart in the body. The information is the blood and MIS is the heart. In the body the heart plays the role of supplying pure blood to all the elements of the body including the brain. The heart work faster and supplies more blood when needed. It regulates and controls the incoming impure blood, processed it and sends it to the destination in the quantity needed. It fulfills the needs of blood supply to human body in normal course and also in crisis.

The MIS plays exactly the same role in the organization. The system ensures that an appropriate data is collected from the various sources, processed and send further to all the needy destinations. The system is expected to fulfill the information needs of an individual, a group of individuals, the management functionaries: the managers and top management.

Here are some of the important roles of the MIS:

- I. The MIS satisfies the diverse needs through variety of systems such as query system, analysis system, modeling system and decision support system.
- II. The MIS helps in strategic planning, management control, operational control and transaction processing. The MIS helps in the clerical personal in the transaction processing and answers the queries on the data pertaining to the transaction, the status of a particular record and reference on a variety of documents.
- III. The MIS helps the junior management personnel by providing the operational data for planning, scheduling and control , and helps them further in decision-making at the operation level to correct an out of control situation.
- IV. The MIS helps the middle management in short term planning, target setting and controlling the business functions. It is supported by the use of the management tools of planning and control.
- V. The MIS helps the top level management in goal setting, strategic planning and evolving the business plans and their implementation.
- VI. The MIS plays the role of information generation, communication, problem identification and helps in the process of decision-making. The MIS, therefore, plays a vital role in the management, administration and operation of an organization.

2.3. Data, Information, Knowledge and Wisdom

There is probably no segment of activity in the world attracting as much attention at present as that of knowledge management. Yet as I entered this arena of activity I quickly found there didn't seem to be a wealth of sources that seemed to make sense in terms of defining what knowledge actually was, and how was it differentiated from data, information, and wisdom. What follows is the current level of understanding I have been able to piece together regarding data, information, knowledge, and wisdom. I figured to understand one of them I had to understand all of them.

According to Russell Ackoff [1989], a systems theorist and professor of organizational change, the content of the human mind can be classified into five categories:

1. **Data:** symbols
2. **Information:** data that are processed to be useful; provides answers to "who", "what", "where", and "when" questions
3. **Knowledge:** application of data and information; answers "how" questions
4. **Understanding:** appreciation of "why"
5. **Wisdom:** evaluated understanding.

Ackoff indicates that the first four categories relate to the past; they deal with what has been or what is known. Only the fifth category, wisdom, deals with the future because it incorporates vision and design. With wisdom, people can create the future rather than just grasp the present and past. But achieving wisdom isn't easy; people must move successively through the other categories.

A further elaboration of Ackoff's definitions follows:

Data... data is raw. It simply exists and has no significance beyond its existence (in and of itself). It can exist in any form, usable or not. It does not have meaning of itself. In computer parlance, a spreadsheet generally starts out by holding data.

Information... information is data that has been given meaning by way of relational connection. This "meaning" can be useful, but does not have to be. In computer parlance, a relational database makes information from the data stored within it.

Knowledge... knowledge is the appropriate collection of information, such that its intent is to be useful. Knowledge is a deterministic process. When someone "memorizes" information (as less-aspiring test-bound students often do), then they have amassed knowledge. This knowledge has useful meaning to them, but it does not provide for, in and of itself, integration such as would infer further knowledge. For example, elementary school children memorize, or amass knowledge of, the "times table". They can tell you

that " $2 \times 2 = 4$ " because they have amassed that knowledge (it being included in the times table). But when asked what is " 1267×300 ", they can not respond correctly because that entry is not in their times table. To correctly answer such a question requires a true cognitive and analytical ability that is only encompassed in the next level... understanding. In computer parlance, most of the applications we use (modeling, simulation, etc.) exercise some type of stored knowledge.

Understanding... understanding is an interpolative and probabilistic process. It is cognitive and analytical. It is the process by which I can take knowledge and synthesize new knowledge from the previously held knowledge. The difference between understanding and knowledge is the difference between "learning" and "memorizing". People who have understanding can undertake useful actions because they can synthesize new knowledge, or in some cases, at least new information, from what is previously known (and understood). That is, understanding can build upon currently held information, knowledge and understanding itself. In computer parlance, AI systems possess understanding in the sense that they are able to synthesize new knowledge from previously stored information and knowledge.

Wisdom... wisdom is an extrapolative and non-deterministic, non-probabilistic process. It calls upon all the previous levels of consciousness, and specifically upon special types of human programming (moral, ethical codes, etc.). It beckons to give us understanding about which there has previously been no understanding, and in doing so, goes far beyond understanding itself. It is the essence of philosophical probing. Unlike the previous four levels, it asks questions to which there is no (easily-achievable) answer, and in some cases, to which there can be no humanly-known answer period. Wisdom is therefore, the process by which we also discern, or judge, between right and wrong, good and bad. I personally believe that computers do not have, and will never have the ability to possess wisdom. Wisdom is a uniquely human state, or as I see it, wisdom requires one to have a soul, for it resides as much in the heart as in the mind. And a soul is something machines will never possess (or perhaps I should reword that to say, a soul is something that, in general, will never possess a machine).

Personally I contend that the sequence is a bit less involved than described by Ackoff. The following diagram represents the transitions from data, to information, to knowledge,

and finally to wisdom, and it is understanding that supports the transition from each stage to the next. Understanding is not a separate level of its own.

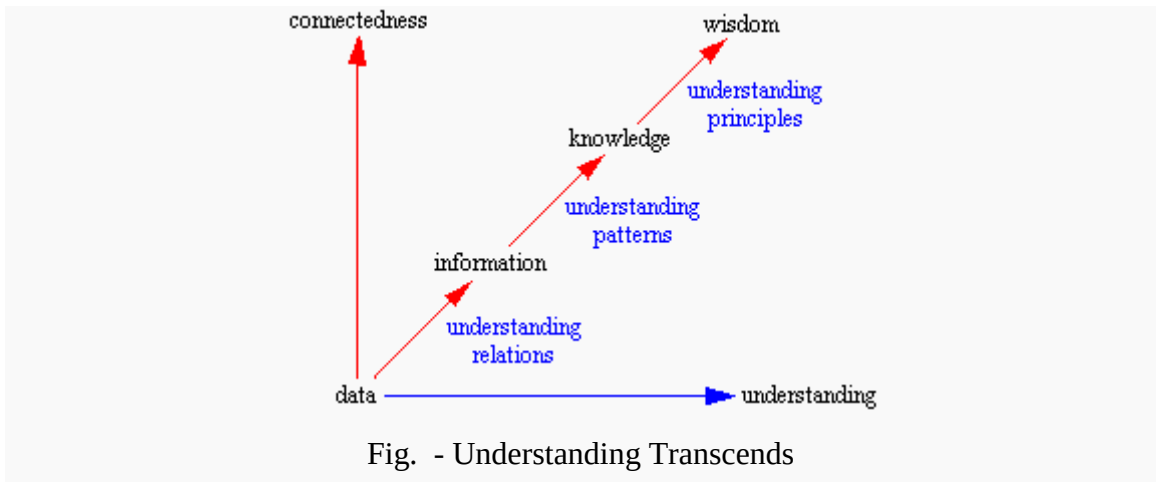


Fig. - Understanding Transcends

Data represents a fact or statement of event without relation to other things.

Ex: It is raining.

Information embodies the understanding of a relationship of some sort, possibly cause and effect.

Ex: The temperature dropped 15 degrees and then it started raining.

Knowledge represents a pattern that connects and generally provides a high level of predictability as to what is described or what will happen next.

Ex: If the humidity is very high and the temperature drops substantially the atmosphere is often unlikely to be able to hold the moisture so it rains.

Wisdom embodies more of an understanding of fundamental principles embodied within the knowledge that are essentially the basis for the knowledge being what it is. Wisdom is essentially systemic.

Ex: It rains because it rains. And this encompasses an understanding of all the interactions that happen between raining, evaporation, air currents, temperature gradients, changes, and raining

2.4. The Information Needs and Sources of Managers

Information is needed for decision making at all levels of management.

Managers at different organizational levels make different types of decisions, control different types of processes, and have different information needs.

Three classical levels of management include:

- Strategic
- Tactical (middle)
- Operational.

Titles have different values in different organizations.

For example, a vice president at a financial organization may not even be a middle manager.

Strategic managers operate in a highly unstructured environment and use EISs, and DSSs. Historically, the most common (organizational structure) was a generic **pyramid** shaped hierarchy with a few leaders at the top and an increasing number of workers at each subsequent lower managerial and operational level.

The pyramid is getting flatter.

In 1993 in the U.S. alone, some 450,000 middle managers lost their jobs.

Some small, knowledge-intensive companies have adopted a **matrix** pattern as their organizational structure, with no one leader and leadership distributed among many more people, varying by project, product, or discipline.

Matrix management includes having multiple bosses.

Technology aside, the politics of information within an organization can undermine optimal business decision making if it is not taken into account when developing systems, and deciding how people will support these systems.

Sub-optimization -- the optimization of an individual or a department at the expense of the larger organization.

In many organizations, clerical and shop floor workers make up the largest group of workers.

Operational managers are responsible for daily operations. They make decisions concerning a narrow time span about the deployment of small groups of clerical and/or shop floor workers.

Middle, or tactical, managers receive strategic decisions from above as general directives. Using those directives as guidelines, they develop tactics to meet those strategic directives. That is, they make decisions concerning how and when specific resources will be utilized. Usually, a middle manager will be responsible for several operational managers.

Responsible for finding the best operational measures to accomplish their superiors' strategic decisions. While a tactical decision concentrates on how to do something, a strategic decision focuses on what to do.

Strategic managers, and directors, that make decisions that affect the entire organization, or large parts of it, and leave an impact in the long run.

People in different management levels have different information needs.

Most of the information that managers require is used to make decisions.

The decision making process of middle managers and above is less structured than that of operational managers;

In general, strategic decisions have no proven methods for selecting a course of action that guarantees a predicted outcome.

Data Characteristics determine where and how the data will be used.

Data range refers to the amount of data from which information is extracted&

Time span refers to how long a period of time the data covers.

Level of detail is the degree to which the information generated is specific.

Internally or externally sourced.

Structured data are numbers and facts that can be conveniently stored and retrieved in an orderly manner.

Unstructured data are drawn from meeting discussions, private conversations, textual documents, graphics, graphical representations, and other non uniform sources.

The higher the manager, the less structured the decisions that a manager faces.

The Web: The Great Equalizer

Managers plan and control.

Planning's main ingredients include:

- Scheduling
- Budgeting
- Resource allocation.

Budget is the most important part of business planning.

The plan is the basis for operations. Control is the activities that ensure operations according to the plan. Both planning and control involve decision making.

A **decision** is a commitment to act. Most of a manager's day is devoted to meetings that produce decisions. Managers control actual activities by comparing actual results to expected results. When discrepancies between the planned and actual performance are found, managers determine the reason for the variance. Management by exception is where a manager only reviews those areas that have deviated from the expected.

Characteristics of Effective Information

Certain types of information can be grasped more quickly when presented graphically.

Many applications allow the user to select how the data will be selected.

Three dimensional graphics are an option now as well.

Dynamic representation usually includes moving images that represent either the speed or direction of what is happening in real time.

Type of Information Systems

POS and other TPS . Clerical and Shop Floor Workers

TPSs are interfaced with applications that provide clerical workers and operational managers with up to date information.

They are also used by operational managers to generate ad hoc reports.

Transaction Processing Systems --- Operational

Decision Support and Expert Systems . Middle Managers

Executive Information Systems (EIS) . Provide managers with timely and concise information about the performance of their organization.

Online analytical processing (OLAP) applications are designed to let a user view a cube of tables showing relationships among several related variables.

Politics is the decision to act in the interest of the individual decision maker rather than in the interest of the organization as a whole

Political tactics include the insistence on adding features that will afford the manager more control, trying to derail the development effort by not cooperating with the developers, and promotion of alternatives to the system.

Enterprise wide systems are shared by many business units and managerial levels.

2.5. A Framework for Information Systems

The ability to share and re-use, to the maximum extent possible, current and future investment is considered a critical element for ensuring the best return on investment as well as the required degree of consolidation. In this respect, one of the key objectives of the Information Systems Framework is to act as an ‘information toolbox’ for information systems to increase their application scope and capabilities by making use of shareable and mashable ‘intelligence’. This augmented intelligence and subsequently value can be provided via investments made in corporate and shared systems. Such approach towards the capitalization of external constituent intelligence by such systems in order to create new and innovative systems able to cater for an even wider set of functionality, is considered to be one of the best ways to introduce further innovation as well as reduce costs.

The Information Systems Framework presents the Government Strategic and Business

Models as the underlying patterns to the approaches Government intends to undertake in order to achieve the objectives highlighted above. It also portrays information Systems Blueprint intended to provide a high level perspective of the key vertical and horizontal systems that constitute the core information system landscape. A number of fundamental technology guiding principles are also described briefly, and mandated where contextually appropriate, to ensure the required degree of interoperability of solutions. These include open standards and service oriented architectures. This also increases the ability to ensure maximum re-use and hence better return on investment.

The Information Systems Framework also provides a basic set of indicators which specify best practices and patterns that can be applied to ensure smooth integration between the various horizontal and vertical solutions. This should enable quicker and additional service orchestrations which introduce the additional value added services from current investments. In this context, it is important that the necessary governance parameters, roles and responsibilities are clearly articulated. The Information Systems Framework provides a set of key guidelines to this extent as well.

This document should be read in conjunction with the National Interoperability Framework (NIF). One of the key objectives of the NIF is to present a strong vision of a “connected” Government through the alignment of departmental business processes, the standardisation, discovery and reuse of ICT assets and a continuous rigour in improving the level of trust in the services provided by the Public Sector. During the articulation of the NIF, the European context was duly considered by incorporating the principles and recommendations of initiatives such as the European Interoperability Framework,

European Interoperability Strategy and the Semantic Interoperability Centre Europe (SEMIC.EU). The building blocks that are proposed are based on international best practices and experience acquired from 20 years of continuous ICT investment. Finally, given the continued investments in the overall Information System landscape, as well as the opportunities to come up with innovative services based on current building blocks, the Information Systems Framework needs to be continuously kept up to date to reflect the current state. It should therefore be treated as a document which has a revision period of 12 months

The primary objectives of the ISF is to describe a number of key building blocks that constitute the primary information system ecosystem currently in use within Government

as well as to provide a clear set of definitions, roles and responsibilities applicable in context. The ISF brings together a number of perspectives including governance, regulatory and business which enable its audience to have concrete visibility of Government's Information System landscape. A set of definitions are also provided, intended to ensure a common baseline for the interpretation of significant information systems concepts as applied within the same ecosystem.

The key audience of this Information Systems Framework includes:

- Ministries & Chief Information Officers (CIO's)
- Private and Public Sector
- General Public

2.6. Business Systems

2.6.1. e-business

E-business is a term used to describe businesses run on the Internet, or utilizing Internet technologies to improve the productivity or profitability of a business. In a more general sense, the term may be used to describe any form of electronic business that is to say, any business which utilizes a computer. This usage is somewhat archaic, however, and in most contexts e-business refers exclusively to Internet businesses.

The most common implementation of e-business is as an additional, or in some cases primary, storefront. By selling products and services online, an e-business is able to reach a much wider consumer base than any traditional brick-and-mortar store could ever hope for. This function of e-business is referred to as ecommerce, and the terms are occasionally used interchangeably.

An e-business may also use the Internet to acquire wholesale products or supplies for in-house production. This facet of e-business is sometimes referred to as e-procurement, and may offer businesses the opportunity to cut their costs dramatically. Even many e-businesses which operate without an electronic storefront now use e-procurement as a way to better track and manage their purchasing.

In addition to buying and selling products, e-business may also handle other traditional business aspects. The use of electronic chat as a form of technical and customer support is an excellent example of this. An e-business which uses chat to supplement its traditional phone support finds a system which saves incredible amounts of time while providing opportunities unavailable through traditional support. By using virtual computer systems, for example, technical support operators can remotely access a

customer's computer and assist them in correcting a problem. And with the download of a small program, all pertinent information about the hardware and software specifications for a user's computer may be relayed to the support operator directly, without having to walk a customer through personally collecting the data.

Using email and private websites as a method for dispensing internal memos and white sheets is another use of the Internet by e-business. Rather than producing time-intensive and costly physical copies for each employee, a central server or email list can serve as an efficient method for distributing necessary information.

In the past few years, virtually all businesses have become, to some degree or another, an e-business. The pervasiveness of Internet technology, readily available solutions, and the repeatedly demonstrated benefits of electronic technology has made e-business the obvious path. This trend continues with new technologies, such as Internet-enabled cell phones and PDAs, and the trend of e-business saturation will most likely continue for some time.

E-business (or electronic business) is the term used to describe using the internet to operate your business. E-business represents only a fraction of worldwide business, but is one of the fastest growing sectors and provides entrepreneurs with excellent opportunities to enter the market.

In a historical sense, the internet is a relatively new way in which to do business, with the growth of e-business taking off in the 1990's. The internet provides consumers with an increasing amount of ways to interact with businesses and has made buying and selling more competitive worldwide. The rate of globalization has been significantly impacted by the internet, making it easier for people and organizations separated by distance to communicate and interact with each other.

E-business can be as simple as using the internet to send emails between staff or communicate with suppliers. A business can be considered an e-business even if it doesn't buy and sell products over the internet, as the term refers to business activities that are assisted by the internet. For example, a successful shop front business may decide to create a website to promote their business but not actually sell products directly through the internet. Almost all businesses are now an e-business to at least some extent; however, many have taken a step further and are involved with e-commerce.

E-commerce specifically refers to ordering, buying, selling and paying for products and services using the internet. E-commerce is a subset of e-business and can make up the entirety of the business or be used alongside existing, traditional business models. E-commerce uses the internet to market, sell and conduct transactions with customers without face to face contact between the buyer and seller. Many e-businesses now conduct all of their operations online and have no physical store that customers are able to visit.

As the internet continues to grow in popularity, technology improves and access becomes more widely available, more and more entrepreneurs are expected to capitalize on the potential of using the internet to operate their business.

The internet and the emergence of e-business has provided entrepreneurs with many new advantages and opportunities. The internet has created a business environment in which time and distance are less important, people have access to more information to help them make decisions and consumers have better access to a broader range of products and services.

A significant benefit for entrepreneurs is that the initial investment for starting up an e-business is generally lower than the costs associated with starting an equivalent business using a traditional model. The technology used to setup and operate an e-business is becoming more advanced whilst also becoming cheaper to obtain. Many successful e-businesses are started using only a home based office with a personal computer, some basic software and a connection to the internet.

Low start-up costs mean that more people are able to enter the market with their business ideas. Many of the barriers to opening your own business have been reduced or removed, with many people able to operate an e-business whilst remaining employed in their regular day job. People who lacked the confidence to start a full scale business are now able to test their abilities and ideas online for a relatively small initial outlay.

The increases in technology and falling prices also provide online businesses with a steady stream of new customers as more people move online to shop for products and services. Cost and access is becoming less of a barrier for consumers and people are becoming increasingly comfortable with making purchases online.

UNIT 3: INFORMATION TECHNOLOGY

✪ Introduction

IT Stands for "Information Technology," and is pronounced "I.T." It refers to anything related to computing technology, such as networking, hardware, software, the Internet, or the people that work with these technologies. Many companies now have IT departments for managing the computers, networks, and other technical areas of their businesses. IT jobs include computer programming, network administration, computer engineering, Web development, technical support, and many other related occupations. Since we live in the "information age," information technology has become a part of our everyday lives. That means the term "IT" already highly overused is here to stay.

3.1. Introduction

We use the term information technology or IT to refer to an entire industry. In actuality, information technology is the use of computers and software to manage information. In some companies, this is referred to as Management Information Services (or MIS) or simply as Information Services (or IS). The information technology department of a large company would be responsible for storing information, protecting information, processing the information, transmitting the information as necessary, and later retrieving information as necessary.

Information technology (IT) is the application of computers and telecommunications equipment to store, retrieve, transmit and manipulate data, often in the context of a business or other enterprise. The term is commonly used as a synonym for computers and computer networks, but it also encompasses other information distribution technologies such as television and telephones. Several industries are associated with information technology, such as computer hardware, software, electronics, semiconductors, internet, telecom equipment, and e-commerce and computer services.

In a business context, the Information Technology Association of America has defined information technology as "the study, design, development, application, implementation, support or management of computer-based information systems". The responsibilities of those working in the field include network administration, software development and

installation, and the planning and management of an organization's technology life cycle, by which hardware and software is maintained, upgraded and replaced.

Humans have been storing, retrieving, manipulating and communicating information since the Sumerians in Mesopotamia developed writing in about 3000 BC, but the term "information technology" in its modern sense first appeared in a 1958 article published in the *Harvard Business Review*; authors Harold J. Leavitt and Thomas L. Whisler commented that "the new technology does not yet have a single established name. We shall call it information technology (IT)." .

3.2. Hardware: Computing, Storing and communicating

Computer Hardware: computer hardware refers to computer systems and other associated equipment including the communication links that a modern IT installation may need. With the rapid advancement in computer hardware technology, a large variety of computer systems with varying sizes, speeds, and functional feature are available. To be able to determine the hardware resource requirements, it is necessary to properly assess the nature of IT needs, the volume of data to be processed, sources of data, complexity of data analysis and impact of other related factors. Such an assessment should also make provisions for expected changes in these factors in the future.

The following list presents computing power from the least expensive and least powerful to the most expensive and most powerful.

- i) Microcomputers (sometimes called personal computers). Based on a single tiny silicon chip called a microprocessor that contains all the essential elements of a computer.
 - Desktops
 - Portables (like laptop, notebook, and palmtop or handheld computers)
- ii) Workstations: faster and more powerful than microcomputers, workstations are used by engineers, architects, scientists, commercial artists, and others who heed computers for speedy number crunching and graphics
- iii) Minicomputers: generally have more speed and power than workstations, but less than mainframes. They can handle several hundred simultaneous users and can run multiple programs concurrently.

-
- iv) Mainframes: are the standards for large business and government agencies, although computer networks composed of smaller computers are threatening to surpass them.
 - v) Supercomputers: are the largest and powerful computers.

The term computer hardware refers to the various electronic components that are required for you to use a computer along with the hardware components inside the computer case. As you know your computer equipment is made of several common components. These include:

The main computer box

A monitor - Looks like a television screen.

- A keyboard.
- A mouse.
- Speakers.
- An optional printer

The main computer box is the main component of the computer. It has computer hardware parts inside that perform the following functions:

Temporary storage of information (known as data in more technical terms) - This function is done by memory.

Permanent storage of information - This function is done by a hard disk, floppy disk, or CD ROM. Manipulation or processing of data - Used to determine where data is stored and perform calculations which support operations that the user is doing.

Interfacing to the outside components or to the outside world - This supports the ability for the user to communicate with the computer and know how the computer is responding to commands which are done primarily through the monitor, keyboard, and mouse along with their interface components in the main computer box. A power supply which provides the electrical power to the components in the computer box

3.3. programs: In Charge of the Hardware

Software programs, computer programs, or computer software, are programs or routines for a computer system or part of the system itself. Application programs are often additions, specially made for the operating system, and designed to add various types of functionality for the user. Software of many different types is made for users with different aims. Categories frequently used for these programs include design and publishing software, games, educational software, music and audio software, productivity software, reference software, personal finance software, utilities, and security software.

Design and publishing software programs provide contexts in which users can create illustrations, edit photographs, and lay out documents for publication. Illustration software includes drawing and painting tools, shapes, and clip art that can be combined and layered with lettering and enhanced with special effects. Peripherals such as a stylus and tablet for entry may give the user finer control for illustration and photograph editing. Photographs can be touched up, cropped, and altered in other ways. Simple and complex documents, including those that combine text and images, can be created with the elements locked into position for certain results.

Software: putting it simply, sets of inter related computer programs are collectively called computer software. A program is a set of instructions, which enable particular processes to be performed by a computer. Software is lifeline of the IT infrastructure and it makes the computer hardware function. Software determines what the hardware does and makes the hardware perform. Software, thus, is a very important IT resource and must be carefully selected and properly maintained. Software, being intangible, sometimes gets a back seat in planning for an IT infrastructure. In fact, improper selection of software may become a major cause of failure of an IT infrastructure in achieving its objectives. There are two types of software:

1. **System software:** programs that control and coordinate the operation of the various types of equipment in a computer system.
2. **Applications software:** programs that allow you to apply the computer to solve a specific problem or perform a specific task (i.e., word processing, spreadsheet, graphics, database programs).

3.4. People: The Most Important Element

The effectiveness of an IT infrastructure is a function of the type of IT personnel available to it. In fact, best of computer hardware, software and data may be spoiled and may cause havoc if the human resources are not competent, honest or reliable. Qualities like competence, honesty and reliability are critical to any work environment. These include computer operators, systems analysts, programmers, data preparation personnel, information systems management, data administrators, etc.

Operations personnel, such as network administrators, are responsible for deploying, configuring, and monitoring the system.

An understanding of the content in the Active Directory protocol documentation is useful to operations personnel because they will need to use many of these protocols to provision information into the system (such as creating and deleting user accounts) and to generate reports about the state of the system (such as verifying that no unexpected accounts are present).

Another advantage of the Active Directory System is that, by providing a centralized store for information used by applications in the enterprise, it reduces the number of distinct information stores that the operations personnel are responsible for maintaining. Instead of having to oversee one information store per application, the operations staff need only oversee a single store that spans all of their organization's applications.

3.5. Procedures: The Way it Goes

Formal operating procedures are physical components because they exist in a physical form such as a manual or instruction booklet. Three major types of procedures are required:

1. User instruction (for users of the application to record data, employ a terminal to enter or retrieve data, or use the result)
2. Instructions for preparation of input by data preparation personnel
3. Operating instructions for computer operations personnel

The procedures play an important role in the smooth and effective utilization of information resource. They also protect the information resource and maintain its quality. Thus, the procedures as element of information system refer to the instructions to users regarding the use of IT infrastructure for normal day-to-day activity and also for handling special situations such as systems failures and crashes. They include user access permissions and disaster recovery procedures as well.

The efficiency of a company's information technology depends directly on the individuals that implement, configure, troubleshoot and operate the technology infrastructure on a day-to-day basis. The company's policies and procedures are the foundation of efficient technology operations: for a company to have an effective information technology infrastructure, it must have technology, people and processes. Many companies focus only on acquiring technology such as firewalls, anti-virus software, intrusion detection systems, etc., yet fail to implement products that ensure that their employees are fully aware of the company's policies and procedures regarding the operation of the technology.

3.6. Information: The Reasons for Using Information Technology

Strictly speaking, Information Technology (IT) has been influencing society for thousands of years. Smoke signals, the abacus, writing, paper, pens and the printing press were all once considered the forefront of IT because those technologies were the latest and greatest in communication.

But today it is widely accepted that the forefront of IT lies in digital communications. As with all historical Information Technology developments, digital communication technology is revolutionary in terms of society, business, defense and culture.

But what is it about digital communications that sets it apart from all other previous technological advances? What is the role that this technology can play in modern society? Firstly, IT acts as a facilitator. It is a medium that permits large volumes of information to be stored, processed or transferred at lightning speed. This means that there is more information at hand to make decisions, maintain relationships, monitor markets or follow movements.

The power of this notion should not be underestimated, it allows people to make decisions based on masses of up to date information. The military, educational institutions and large business have long understood the power of this notion but it has only been in the last ten years that mainstream society has also embraced the power of information for both social movements and small business.

Secondly, Information Technology is an enabler. Only a select minority of specialized companies in the world can claim that IT is their business. Most corporations and institutions use Information Technology to enable functions that drive their core business. This may include better communication between staff, suppliers or customers or it may include better asset management. Whatever the function, digital communication makes the storage, processing or transfer of information more efficient than ever before. But it is important to note that in essence it does not change core businesses, ideals, ethics or movements. It just enables them to function at a greater scale much more efficiently.

Lastly, Information Technology assumes the role of custodian. As data and information is amassed on just about any subject you care to imagine, massive stores of data are emerging ready to service any new requests. And therein lies the security paradox that modern Information Technology presents. One can argue that more available information makes our lives more efficient, safer and richer. Whilst the other argument is that the

very same information that provides these benefits is also our Achilles heels due to the security implications of unauthorized infiltration.

There is little doubt that the role of information technology is directed and constrained by the very technology that defines it. Today, that technology allows us to collect, manipulate and communicate masses of information. As a consequence, the role of IT revolves around improving the efficiency, capacity, speed or accuracy of any elements that can benefit from the functions that modern Information Technology systems provide.

Information Technology (IT) has long been viewed as having great potential for improving decision making in agriculture. IT has connected the world globally and is now changing our life style and social consciousness dynamically. In all phases of the agricultural industry, information technologies are vital to the management and success of a business. Agriculture has also been greatly influenced by IT.

Information Technology is rapidly becoming more and more visible in society and agriculture. IT refers to how we use information, how we compute information, and how we communicate information to people. People must have computer and information technology. To participate and make informed decisions in the agricultural industry a person must have ability to gather, process, and manipulate data.

3.7. Communication Technologies

Information and Communications Technology or ICT, is often used as an extended synonym for information technology (IT), but is a more specific term that stresses the role of unified and the integration of telecommunications (telephone lines and wireless signals), computers as well as necessary enterprise software, middleware, storage, and audio-visual systems, which enable users to access, store, transmit, and manipulate information.

The phrase *ICT* had been used by academic researchers since the 1980s, but it became popular after it was used in a report to the UK government by Dennis Stevenson in 1997 and in the revised National Curriculum for England, Wales and Northern Ireland in 2000. The term *ICT* is now also used to refer to the convergence of audio-visual and telephone networks with computer networks through a single cabling or link system. There are large economic incentives (huge cost savings due to elimination of the telephone network) to merge the audio-visual, building management and telephone network with the computer

network system using a single unified system of cabling, signal distribution and management.

The term Info communications is sometimes used interchangeably with ICT. In fact Info communications is the expansion of telecommunications with information processing and content handling functions on a common digital technology base.

The introduction and rapid spread of Information and Communications Technologies (ICTs), such as the World Wide Web, e-mail, telephones, fiber optics and satellites is revolutionizing the way in which societies interact, conduct their businesses, compete in international markets and set their economic and human development agendas.

ICTs enable societies to produce, access, adapt and apply information in greater amounts, more rapidly and at reduced costs and offer enormous opportunities for enhancing business and economic viability.

ICTs can also contribute towards strengthening democracy, increasing social participation, competing in the global market place and removing barriers to modernization; making poor populations fuller agents in the sustainable developmental process.

However, with the rapid introduction of these technologies in the western world, the gap between the rich and the poor nations is widening. A 'global information' society threatens to create a new divide between the 'information rich' and 'information poor' people of the world as poorer and marginalized people in developing countries are excluded from the benefits of new communication technologies.

3.8. Database Management (DBM)

Database management involves the monitoring, administration, and maintenance of the databases and database groups in your enterprise. Enterprise Manager is the premier tool for managing your database environment.

With Enterprise Manager, you receive:

- A complete set of integrated features for managing Oracle Databases
- Unparalleled scalability that lets you manage a single database or thousands of instances
- An intuitive management product that leads the industry in ease of deployment and use

-
- A database management system (DBMS) is simply the software that permits an organization, to centralize data, manage them efficiently, and provide access to the stored data by application programs. Consider the following figure that illustrates the database concept.

The DBMS acts as an interface between application programs and the physical data files. When the application program calls for data item such as gross pay, the DBMS finds this item in the database and presents it to the application program. Using traditional data files, the programmer would have to define the data and then tell the computer when they are. The following figure illustrates the elements of a database management system.

A database management system has three components:

- A data definition language
 - A data manipulation language
 - A data directory
- **Data definition language:** the data definition language is the formal language used by programmers to structure of the database. The data definition language defines each data element as it appears in the database before that data element is translated into the forms required by the application programs.
 - **Data manipulation language:** most DBMSs have a specialized language called a data manipulation language that is used in conjunction with some conventional third or fourth generation programming languages to manipulate the data in the database. This language contains commands that permit end users and programming specialists to extract data from the database to satisfy information requests and develop applications. The most prominent data manipulation language today is SQL, or structured Query language.
 - **Data dictionary:** the third element of DBMSs is a data dictionary. This is an automated or manual file that stores definitions of data elements and data characteristics such as usage, physical representation, ownership (who in the organization is responsible for maintaining the data), authorization, and security. Many data dictionaries can produce lists and reports of data utilization, groupings, program locations, and so on.

Advantages of Database Management System

The advantages of a DBMS are as follows:

- Complexity of the organization's information system environment can be reduced by central management of data, access, utilization, and security

-
- Data redundancy and inconsistency can be reduced by eliminating all of the isolated files in which the same data elements are repeated
 - Data confusion can be eliminated by providing central control of data creation and definitions
 - Program data dependence can be reduced by separating the logical view of data from its physical arrangement
 - Program development and maintenance costs can be radically reduced
 - Flexibility of information systems can be greatly enhanced/by permitting rapid and inexpensive ad hoc queries of very large pools of information
 - Access and availability of information can be increased

UNIT 4: COMMON BUSINESS APPLICATIONS OF INFORMATION TECHNOLOGY

✪ Introduction

The term “*information*” has a diversity of meanings, from everyday usage to technical interpretations. Generally speaking, the concept of information is associated with knowledge derived from study, experience, or instruction. *Technology*, on the other hand, refers to the application of knowledge to the practical aims of human life, or to changing and manipulating the human environment. Technology includes the use of materials, tools, techniques and sources of power to make life easier or more pleasant and work more productive. Technology began to influence human endeavour as soon as people began using tools. Technology also started being used for managing information when the amount and variety of information grew to such vast proportions that the human brain could neither store nor process it efficiently.

Every day, people use computers in new ways. Computers and other electronic devices are becoming increasingly affordable. They continue to be more powerful as information-processing tools as well as easier to use. Humans are continually becoming dependant on IT-enabled devices for carrying out simple tasks like remembering a phone number to complex ones like flying a fighter plane. Information Technology has applications in almost all aspects of our life. Some of the important ones are: science and engineering, business & commerce, education governance, medicine and entertainment

Unit Objectives: After studying this unit, you should be able to:

- Understand the significance of information system for financial management
- Recognize the relationship between marketing and information system
- Introduced to manufacturing and production information systems
- Know human resource information systems
- Understand transaction processing system

4.1. Financial Information System

Whether you are the chief operating office of a large international organization or merely the supervisor of a small department at one location in that organization, you will find it

necessary to make financial decisions. The financial decisions you may encounter as a manager are diverse and could include decisions similar to these;

1. Should you purchase a large new piece of equipment or lease the equipment for a three- year period?
2. How much of your department's funds should you allocate to telephone usage rather than travel expenses?
3. How much does your division have on hand at various times during the year and is it enough to cover the cash expenditures you expect to incur at those times?
4. Should you invest money in new computer equipment or additional merchandise for resale?
5. How will a 20% utility rate increase affect your budget for the coming year?
6. Should you allow a particular customer to make a large purchase on credit?
7. Should your organization decide to offer a new product or service, or should it provide the same products and services in additional markets?

To make these and similar decisions, you need to understand the basic financial accounting systems found in most organizations. You must also understand how financial information systems can help you make improve decisions about the financial problems with in your purview.

The Financial Management Function

The financial management functions of any organization encompass a number of important responsibilities. These include monitoring and analyzing the organization's financial condition through the management of accounting systems and the preparation of financial statements and reports. The financial management function is also responsible for the budgeting process, the management of customer credit, the calculation and payment of income, payroll, property, excise and other taxes; and the management of appropriate insurance coverage for the organization's personnel and assets. Furthermore, the function must provide for an auditing process to ensure the accuracy of the financial information kept and to protect the investor's funds.

The financial management function is also responsible for the management of the organization's fixed assets, pension funds, and investment in existing plants and equipment, subsidiaries, or other investments. The function also includes the evaluation of new investments and the acquisition of borrowed funds or capital to pay for them.

The financial management function must manage the cash flow of the organization in such a ways as to ensure that the organization has the necessary funds to operate, that extra cash is not left idle, and that the organization has the borrowing power to meet its cash needs when the cash flow is insufficient.

Clearly, the financial management function is very important to the organization. Because of its importance, financial management personnel require information systems that provide them with accurate, timely and appropriate information to carry out their functions.

Like all information systems, financial information systems may be categorized into those that support operational tasks, those that support tactical decisions, and those that support strategic planning decisions.

4.1.1. Operational Financial Information Systems

Typically, the applications first computerized by many organizations are accounting systems designed to produce operational level output. The development of computerized financial information systems for use in tactical decision making and strategic planning is usually deferred until the basic operational level information systems are in place and working.

Operational financial information systems produce the routine, repetitive information products that every organization finds necessary. These products, or output, include paychecks, checks to vendor, customer invoices, purchase orders, stocks reports, and other regular forms and reports. Financial operating information systems are typically task oriented. They focus on processing financial transactions to produce the financial products just described.

Accounting Systems

The heart of an organization's financial operating information system is the accounting system. A computerized accounting system is composed of a series of software modules or subsystems that may be used separately or in an integrated fashion. The subsystem modules include:

1. general ledger
2. fixed assets
3. accounts receivable (credit limit, aged accounts receivable reports can be generated)

-
4. order entry
 5. accounts payable
 6. inventory
 7. purchase order
 8. payroll

When these computerized accounting subsystems are integrated, each subsystem receives data as input from other subsystems and provides information as output to other subsystems. The accounting subsystems might be integrated.

4.1.2. Tactical Financial Information Systems

The computerization of accounting systems usually changes the way managers' view accounting information. Because a large database of information becomes available in computerized form and can easily be extracted or manipulated, this information begins to be viewed as a resource for tactical planning. Suddenly, it becomes possible for managers to get important summaries and comparisons of accounting data easily and swiftly. In the past this information would have taken a great deal of time to extract from a manual accounting system. The result is that managers view the accounting system as more than merely a producer of checks, invoices, and statements. It becomes repository of important data that can assist management in decision-making. The computerization of accounting system has helped to spawn the use by managers of corporate databases to support tactical decisions and strategic planning. This has led to the development of computer supported financial information systems for tactical decision making and strategic planning information systems.

The Nature of Financial Information Systems for Tactical Decision Making

Tactical financial information systems support management decision making by providing managers with regular summary reports, regular exception reports, ad hoc reports, and other information that will help them

- 1) control their areas of responsibility and
- 2) Deploy their resources to pursue organization goals. Whereas operational control systems are focused on tasks, tactical information systems are focused on resource allocation.

It is possible to design many computer supported tactical level information systems for the financial decisions that managers must make. Common systems include budgeting

systems, cash management systems, capital budgeting systems, investment management systems. Each of these systems will be briefly described.

Budgeting Systems

The general ledger subsystems of many computerized accounting systems permit the user to enter budget amounts by account number. Periodically, (weekly, monthly, quarterly, or annually) the budgeted amounts (allocations) and the actual amounts spent or received (actual) for each account are compared and various reports are prepared. For example, the general ledger subsystem of a mainframe accounting system may provide the reports listed below.

1. Current budget allocations by line item.
2. Current budget allocations compared to year to date revenues and expenditures.
3. Budget variances by line item type, or the differences between allocations and actual
4. Current budget allocations compared to the previous year's allocations
5. Current revenues and expenditures compared to the previous year's revenues and expenditures
6. Current revenues and expenditures compared to the average of the other units or divisions of the organization.

Reports such as these may be prepared for a department, a division, a subsidiary, or the entire organization. The budgeting system permits managers to compare revenue and expense data against the standard of the budget allocations. It also allows prior fiscal period, other division or department, industry wide data to be used as standards against which current budget amounts may be compared.

Cash Management Systems

Important functions of financial management include ensuring that the organization has sufficient cash to meet its needs, putting excess funds form any period to use through investments, and providing borrowing power to meet the organization's cash needs in those periods when there is an insufficient cash flow.

There are two major reasons why an organization needs cash: for working capital (cash needed for day to day operations) and for the acquisition of long term assets. To determine if adequate cash is available for its working capital needs and its long term

asset acquisition plans, the organization must prepare a report of its expected cash flow for the time periods being considered. Typically this report shows the cash flow for each month of the coming year.

A cash flow report shows the estimated amount of cash that will be received and spent each month. The report shows in which months there will be cash received and spent each month. The report shows in which months there will be excess funds that might be put to use, and in which months there will be insufficient funds, required the organization to borrow cash to meet its working capital or fixed asset acquisition needs.

The information provided by a cash flow helps the manager make decisions about investing, purchasing, and borrowing money. If this information is placed on an electronic spreadsheet, the manager may stimulate a number of possible business conditions, such as (1) increasing or decreasing revenue, (2) increasing or decreasing customer credit problems, (3) deferring the acquisition of an asset, or (4) repairing existing fixed assets instead of replacing them.

Capital Budgeting Systems

A capital budget contains information about the planned acquisition or disposal of major plant assets during the current year. The manager may wish to compare the various capital spending plans in terms of three commonly used evaluation tools: net present value, internal rate of return, and payback period. These tools can easily be calculated using computer systems. “What if?” games are performed using software packages

Investment Management Systems

Investment management – overseeing the organization’s investments in stocks, bonds, and other securities – is an important part of cash management. Managing investment is also an important part of an organization’s pension plan management. Whatever the source of the funds for investments, most organizations invest money in securities of one kind or another. Careful management of these investments is necessary to ensure the achievements of organization goals.

Current information systems provide unique ways to manage stock and bond portfolios. These ways typically involve the use of external databases that furnish immediate updating of stock and bond prices, information about the history of each investment, and various portfolio investment analysis tools to help the manager stay on top of the

organization's investments. The system may be a simple one in which the manager's microcomputer is equipped with a modem and he or she is provided with a subscription to an investment service. Possible information that can be generated are current dividend, price history, price stability index, projected changes earnings, current earnings per share, debt as a percent of capital, dividends history, industry ranking, low price per year and the like. Tactical financial information systems give the manager increased control over the financial resources of a department or an entire organization, and provide considerable support when he or she is deciding the allocation of financial resources to meet organizational goals.

4.1.3. Strategic Financial Information Systems

You have learned that operational control level information systems are task oriented and that tactical level information systems are resource allocation oriented. In contrast, strategic information systems are goal oriented. That is, these systems are concerned about goal and direction setting for organizations.

Strategic information systems typically include several types of information flows:

1. Financial condition analysis data or an analysis of internally generated information describing the status of the organization
2. Economic, demographic, and social data or an analysis of externally generated data describing the present and future environment for the organization
3. Forecasts of the future of that organization in those environments

Two major outcomes of financial strategic planning are the setting of financial goals and directions for the organization. The former may include setting investment goals and return on investment goals. The latter may involve deciding on new investment opportunities or on the mix of capital sources used to fund the organization.

A major source of computerized information about the current and future status of the organization is the organization's own financial accounting database. A promising source of computerized information pertaining to the present and future environment in which the organization must operate are on line databases for economic, social, demographic, technological, and political information. Projecting likely scenarios for the organization using these two categories of data is the art of forecasting. A major purpose of strategic decision-making is to use long-range forecasts to reduce the risk involved in major organization decisions.

Financial Conditions Analysis Systems

Computerized accounting systems provide the user with a variety of reports on which many ratios and analysis tools may be applied.

Example, working capital = current assets – current liabilities.

Long Range Forecasting Systems

Strategic planners demand forecasts on a variety of factors that will affect organization performance in the future.

- Forecasting the sales revenue of a potential new product makes important information available to planners considering the development and marketing of the new the product.
- Forecasting the financial health of the organization through the long range budget estimates – including a variety of possible wage negotiation settlement, actions by competitors, interest rate fluctuations, fuel costs changes, and different inflation rates provide planners with opportunities to consider actions that will help the organization survive bad times or take advantage of a future environment.

The types of information used in forecasting the future environment of the organization are many and diverse. They include descriptions of the present economy and forecasts of the future economy, information on the present demographic structure of the region or country and forecasts of the demographic structure of the region or country, and descriptions of the current social structure and social mores and predictions on the future structure of society and societal mores.

4.2. Marketing Information Systems

Marketing information systems support the major activities of a marketing organization. The information systems collect data that describe marketing operations, process those data, and make marketing information available to marketing managers to help them make effective decisions.

4.2.1. Operational Marketing Information Systems

Operational marketing information systems primarily produced routine, repetitive, expected, and regular data that describe past marketing activities. The information they produced is usually detailed, highly structured, and accurate, and is derived from internal sources.

Sales Information Systems

Salespeople are responsible for many sales activities. They must identify potential customers, make the customer contact, call on the customer, make the sales, close the sale, and follow up on the sale. These are the bread-butter sales activities of the salesperson. There are many information systems that support the salesperson in these activities. Among these information systems are the following:

Prospect Information System

Locating potential customers is often a time consuming and frustrating part of the salesperson's work. The sources of information used to obtain leads about prospective customers are frequently diverse and may include other customers, other vendors who sell supporting or ancillary products, newspapers notices, telephone directories, and direct customer inquiries. The search of directories and other lists of customers may be very time consuming and yield few customers.

Files of sales leads are often called prospect files. When these files are stored on magnetic media, they are easier for the salesperson to search or summarize. Outputs of prospect information systems may include lists of prospects by location, by product category, by gross revenue, or by other classifications important to the sales force.

Contact Information Systems

Customer contact information systems provide information to the sales force pertaining to customers, their product or service preferences, sales history data, and a historical record of sales calls and/or visits. One output of these systems may be a call report showing the number of sales calls made by a salesperson categorized by size of organization, previous sales, or some other characteristics, and the number of sales made per customer, per visit, and or per category.

Inquiry Information Systems

When customers make inquiries about the products and services the organization offers, the inquiries need to be processed, recorded, and stored for analysis or for sales contact. It is important that inquiries be associated with the actual or potential customer who made the inquiry, what products or services the query pertained to, when the inquiry was made, and where the potential customer was located. It is also very important to record these data on a medium that will allow analysis easily at some future time.

Document Information Systems

- Sales form letters for salespeople so that they can mail well written, clear and effective letters
- Quotation documents are frequently produced. A document information system provides marketing personnel with ready-to-adapt documents for their operations. The system also improves the quality of the documents developed by marketing personnel, and is likely to increase sales revenue.

Order Entry Systems

A great deal of marketing data is created through the financial accounting system of the organization. For example, the order entry system provides the marketing manager with the raw data from which to obtain gross sales by time period, salesperson, product, and territory. This information may be used for a variety of marketing decisions at several marketing decision levels. Analysis of product sales through analysis of the sales orders provides the marketing managers with some of the information on which sales forecasts can be based, for instance.

Telemarketing Systems

Using the telephone for selling is referred to as telemarketing. Using telemarketing has become a very important means by which companies have improved the productivity of their sales force. Using the telephone to initiate contacts, offer products and services, follow up on sales eliminates travel costs and travel time and lets salespeople reach many more customers in a given time period than they could have through conventional means.

Some telemarketing systems include computer support for the automatic calling of parties and/or the automatic delivering of a voice message to the answering party under the control of computer system.

Direct Mail Advertising Systems

Many firms generate sales by mailing sales brochures and catalogs directly to customers. To distribute sales documents rapidly to large numbers of potential customers, most marketing departments maintain customer mailing lists that are used for mass mailings. The lists may be drawn from customer files, accounts receivable records, or prospect files, or they can be purchased from other documents.

Inventory Information Systems

Inventory subsystems of the financial accounting system provide information about inventory levels, stock outs conditions, stock receipts, stock issues, stock damage, and the location and distribution of stock within the organization. For example, salespeople should be able to check the amount of available stock in inventory before closing a sale with a customer. Thus, the inventory information system provides important data regarding stock levels to the order entry system.

Distribution Information Systems

If the organization uses its own distribution system, information about its effectiveness must be collected and reported to management. Information should also be maintained about the acquisition, repair, use and allocation of equipment.

Credit Information Systems

An operational level credit information system usually provides the salesperson or credit manager with information pertaining to the maximum allowable credit that should be granted to a customer. This information is ordinarily integrated with the order entry subsystem of the financial accounting system. The credit information system may manage information on the credit history of customers; this data can include information from external sources, such as credit agencies.

4.2.2. Tactical Marketing Information Systems

Tactical information systems differ from operational marketing information systems in that they may provide produce information not only on a regular basis but also generate ad hoc reports, create unexpected as well as expected output, produce comparative as well as descriptive information, provide summary information as opposed to detailed data, included both internal and external data sources, and processes subjective as well as objective data.

Examples of tactical marketing information systems that will be discussed in this part of the paper include those that support the development and management of the sales force, the management of advertising and promotion campaigns, the distribution and delivery of products sold, and product pricing.

Sales Management Information System

A major objective of sales manager is to reach the sales goals set by top management. To accomplish this objective, sales managers must make many tactical decisions, such as

sales territories should be shaped, how the sales force should be allocated within those territories, and what emphasis should be placed on the products offered and customers served. Sales managers must decide how to reward salespeople to encourage increased sales efforts, which market segments should be emphasized to best reach sales goals, and which products and services will best appeal to each segment. They also must monitor the progress of the sales effort to determine if their decisions were correct or if they need to change their tactical plans.

To make these decisions effectively, sales managers should have at their disposal a great deal of data about the sales histories of each salesperson, territory, product, and market segment. Managers can use these data – provided by sales management information systems – to develop reports analyzing sales activities that help them make decisions about salespeople, territories, products, and customers. Analysis of past sales efforts might reveal, for instance that the greatest volume of sales is obtained when certain products are emphasized with certain market segments. This information may be obtained from a report that correlates product or service categories with customer categories.

Through the use of these reports, marketing managers may adjust advertising media, promotion schemes, sales calls, and sales approaches to emphasize selected products to very narrowly defined market segments to maximize gross sales.

In addition to planning future campaigns, sales managers need information to control current campaigns. The manager monitoring sales progress may wish to receive daily, weekly, or monthly reports on the sales achievement of each salesperson with averages calculated for the entire sales force. This information might be used to evaluate salesperson effectiveness, the success of the sales reward system, or the appropriateness of the shape of the territories assigned. A report showing the percentage of each salesperson's sales by market segment and by product along with averages for the total sales force would display the current effectiveness of each salesperson in selling the organization's products to each customer group.

The manager can compare sales, product, customer, and territory information from one department against external benchmarks of success, such as the organization as a whole or industry averages. The manager, for example, may wish to compare staff deployment to the deployment of marketing staffs in other marketing departments within the

organization or to the industry as a whole. The marketing manager might also view the amount of money spend in the department on salespeople versus support staff and then compare those amounts to the amounts spend by other marketing departments in the organization or to the industry as a whole. If sales data are added to expense data, the marketing manager can compare the productivity of the industry. Comparisons that indicate poor productivity might lead the marketing manager to compare the productivity of each salesperson to the average for the department. This comparison, in turn, might lead to decisions changing the configuration of sales territories, training for specific salespeople, product emphasis for certain territories, market segment emphasis for certain sales personnel, or a variety of other remedies.

Advertising and promotion information systems

Advertising and promotional tactics also need to be developed by marketing managers to implement strategic sales goals set by top management. Managers must decide which advertising media and promotional devices to use to reach the selected market segments, when these media and devices should be used, and what overall mix of promotional activities should be deployed to achieve sales goals. Advertising and promotion information systems assist managers in these tasks.

To decide what advertising and promotional tactics to use, marketing managers need such information as market segment history, the effectiveness of previous advertising and promotional efforts on each market segment, and the sales history of products by market segment. Historical data on the effectiveness of various advertising and promotional instruments and the sales of products by market segment are not foolproof methods for forecasting future success. Past profiles and trends may not necessarily be projected into the future. The areas of advertising and promotion contain too much managerial art for projection of past data to be perfectly successful. However, the past success of specific advertising and promotion campaigns for specific products and customers is very important information for the manager.

Production Pricing Information Systems

Product pricing information systems are important because the price of a product or service affects the sales volume and profitability of the organization. While the marketing manager usually selects a price that will at least recover production costs, the price chosen is constrained by the prices of competitors for similar products or services and for alternative products or services. To make pricing decisions, the marketing manager

should have information pertaining to the expected demand for the product or similar products, the desired profit margin for the organization, the costs of producing the product or providing the service, and the prices of competing products.

To help marketing manager price products and services, some organizations develop a pricing model that allows him or her to input data on a variety of forces affecting prices such as expected competitive prices, expected consumer price indexes, expected consumer disposable income, volume of expected advertising expenditures. The model then uses arithmetic algorithms that represent the organization's assumptions about the interrelationship of each of these factors. This makes it possible for the manager to "play" with the inputs to identify the recommended price for a product under a variety of possible conditions.

Distribution Channel Decision Support Systems

Deciding how the organization's products and services will be delivered to the ultimate consumer is another important tactical decision that must be made. The organization needs to determine if it wishes to sell all or some of its products and services directly to its customers, use middlemen, or both. If direct sales are considered, then the choice of trade channels or channels within that delivery mode – for example, sales force, direct mail, telemarketing, and so on – must be decided on. If middlemen are chosen to distribute the product or service, the choice of the channel will be affected by how well the channel distributes the products or services among the organization's targeted market segments.

To support the marketing manager, the marketing information system should provide a distribution channel decision support systems. This system should provide information such as the costs of using the various distribution channels, the time lags caused by the various channels, the reliability of the various channels in delivering the products and service, and the marketing segment saturation provided by the channels.

4.2.3. Strategic Marketing Information Systems

To develop an overall marketing plan, an organization may engage in a variety of tactical and strategic planning activities. The strategic activities include segmenting the market into target groups of potential customers based on common characteristics or needs or

wants, selecting those market segments the organization wishes to reach, planning products and services to meet those customers' needs, and forecasting sales for the market segments and products.

Sales Forecasting Information System

Strategic sales forecasting usually includes several varieties of forecasts: forecasts of sales for the entire organization, forecasts of sales for each product or service, and forecasts of sales for a new product or service. The results of these sales forecasts will often be further categorized by sales territory and sales division. Regardless of the type of forecasts, sales forecasts are usually based on more than historical data; they are not merely projections of past trends. Sales forecasts are also based on assumptions about the activities of the competition, government action, shifting customer demand, demographic changes and movements, and a variety of other pertinent factors, including even the weather. Errors in sales forecasting will have many implications in the other aspects of the organization.

Product Planning and Development Information Systems

The major objective of product planning and development information systems is to make information about consumer preferences obtained from the marketing research system available for the development of new products. The primary output of planning and development activities is a set of product specification. The specifications will serve as inputs for other decisions such as product design.

4.2.4. Tactical and Strategic Marketing Information Systems

Two important information gathering systems provide support for both tactical and strategic marketing decisions. These two information systems are marketing research systems and information systems that collect data about the organization's competitors.

Marketing Research

In large organizations, research departments conduct and manage marketing research. In small companies, marketing research may be completed by outside consultants or by personnel who must wear several hats. Regardless of how the function is completed, the results of marketing research provide important inputs to tactical and strategic decision making.

Inputs to marketing research are heavily derived from sources external to the organization. These inputs include such widely diverse sources of customers as

customers, potential customers, census and demographic data, industry or trade data, economic data, social trend data, environmental data, and scientific and technological data. These data may be obtained through such means as direct mail surveys of customers, personal and telephone interviews of consumers, library searches of governmental and industry reports, searches of the databases of information utilities, and reports filed by sales personnel.

Marketing research personnel make heavy use of statistical methodology in analyzing the data collected and in reporting the information to the organization. Obtaining totals, counts, and averages in terms of consumer responses to questions, correlating social and economic characteristics of customers with their buying practices, completing times series analysis of past industry wide sales to determine the projected sales of a product, and testing hypothesis about consumer response to differing product packaging represent only some of the statistical procedures that are used to analyze information for marketing managers.

Typical of the activities of a marketing research department are:

1. Conducting trend analysis of the sales of products and services identical or similar to those offered by the organization to identify products or services that are on the ascent or descent
2. Analyzing population and target group characteristics, especially for trends or changes in data that could affect the organization
3. Analyzing and identifying consumer preferences, including the testing of products and services
4. Determining and analyzing customer satisfaction with the organization's existing products and services
5. Estimating market share for all or each product and service offered.

The results of marketing research are often presented graphically, in the form of tables, charts, and graphs.

Competitive Tracking Information Systems

To ensure that the marketing mix offered by your organization will continue to be effective in satisfying customers, you must keep abreast of major competitors and their marketing activities. Market share is likely, in the end, to be the greatest for the organization that provides the marketing mix most closely matching a given market

segment's needs and wants. Thus, knowledge of competitor prices, products, sales, advertising, and promotions must be gathered if the organization is to avoid falling behind the competition in the eyes of the customers. This task is carried out through competitive tracking information systems.

Information about competitor activities is also obtained from informal sources, through such activities as reading trade journals and newspapers, visiting competitor distribution outlets, and talking to competitor offers and employees at conventions. Salespeople in most organizations are encouraged to provide feedback about competitor activities by filing field reports. The information they obtain may be gathered through observation of competing salespeople or by seemingly casual questions that addressed to customers. Information about the competition may also be gathered more systematically by conducting keyword searches in external databases or information utilities.

4.3. Manufacturing and Production Information Systems

Purpose: manufacturing systems encompasses all the activities necessary to ensure production. These activities may include the evaluation of sites for production; the planning, development, and maintenance of production facilities; and the setting of production goals to meet the requirements of the sales forecast generated by the marketing system.

Manufacturing and Production Information Systems

Manufacturing and production information systems provide the data necessary to plan, organize, operate, monitor, control and otherwise manage production systems.

4.3.1. Operational Production Information Systems

There are numerous operational production information systems. Many are part of the financial accounting system of organization. For example, purchasing, accounts payable, inventory, order entry, accounts receivable, and payroll subsystems of the accounting system provide information to support manufacturing and production activities. The next section briefly describes some major operational production information systems.

Purchasing Information Systems

To produce goods and services, you must have the right quantity of raw materials and production supplies on hand. Furthermore, you will want to procure these materials and supplies at the lowest cost and have them delivered at the right time. To assist in this

function, the purchasing information system has to maintain data on all phases of the acquisition of raw materials and purchased parts used in production.

Receiving Information Systems

When shipment of purchased goods and supplies are received, they must usually be inspected and verified and the information about their status passed on to the accounts payable, inventory, and production departments. Delivery dates should also be noted so that data on delivery times can be collected. This type of information is supplied by receiving information systems.

Quality Control Information Systems

Quality control information systems provide information about the status or production of goods as they move from the raw material state, through goods in process, to the finished goods inventory.

Shipping Information System

At the other end of the production process, there is a need to place finished goods produced into inventory and to ship those on order to customers. There are many records and documents that are used to assist and monitor in the inventorying and shipping processes – for example, shipping reports and packaging slips. The information from the shipping system affects the inventory and accounts receivable systems.

Cost Accounting Information Systems

A variety of operational information subsystems of the financial accounting system collect and report information about the resources that are used in the production processes so that accurate costs of production can be obtained on products and services. Cost accounting systems monitor the three major resources used in production: personnel, materials, and equipment and facilities. Payroll information systems and materials information systems provide information about the costs of these resources.

4.3.2. Tactical Manufacturing and Production Information Systems

Tactical information systems include inventory management and control systems, capacity planning, production scheduling, and product design and development.

Inventory Management and Control Information Systems

The management and control of raw materials, goods in process, and finished goods inventories is an important part of production system. Careful management and control of these inventories will usually provide considerable savings to the organization. Inventory

management and control systems use information from operational information systems, such as the shipping and receiving systems, purchasing systems, and order entry systems.

Capacity Planning Information Systems

In addition to ensuring that there will be enough raw materials on hand for forecasted production, the production manager must also see to it that there will be enough production capacity available to meet production goals. The purpose of capacity planning is to make certain that there is sufficient personnel, space, machines, and other production facilities available at the right time to meet the organization's planned production. Managers also utilize capacity to minimize capacity in excess of planned production needs.

Production Scheduling Information Systems

The purpose of the production schedule is to allocate the use of specific production facilities for the production of finished goods to meet current or forecasted orders. To manage the scheduling process, a number of scheduling tools have been developed. Two of these are Gantt and PERT charts and there are project software packages at this time.

Product Design and Development Information Systems

Numerous tactical decisions must be made to design and develop a product, especially a new product. The design engineering team usually depends on product specification information derived from customer surveys, target population analysis, or other marketing research information systems. The team also depends on data from the quality control information systems. Teams may use other computerized systems for designing new products as well. Software used for design, testing and manufacture of product is commonly available these days.

4.3.3. Strategic Planning Manufacturing Information Systems

As we just discussed, production information systems are primarily operational and tactical in nature. They are concerned with providing information to monitor and control the production of goods and services and to allocate resources to complete production processes. Manufacturing information systems are typically strategic in nature.

For example, the decision to construct a plant addition or a totally new plant, the selection of the plant site, and the creation of general plans for the design and layout of the facility are top management decisions. Decisions of this magnitude will require the

commitment of a large amount of capital and other resources over a long period of time and thus are strategic planning decisions. Such decisions are not made lightly.

The decision to locate a plant site requires a great deal of information that is external to the organization. Facilities planning information systems support top management decisions in this area. Some of the information needed is relatively quantitative – for example, the availability and cost of trained or experienced labor and the degree to which it is unionized, the availability and cost of transportation for raw materials and finished goods, the availability of suitable sites and the cost of land, the proximity of raw materials suppliers and/or finished goods customers, the availability and costs of power, and the rate of property and incoming taxation.

Other information used in locating a plant may be qualitative in nature – for example, community attitudes toward an organization of the type of wishing to locate there and the quality of community services, such as education and training opportunities.

The final decision concerning plant may be made using some form of weighted average technique, in which factors such as those listed above are ranked and the total scored for several potential sites computed. It may also be made on largely emotional factors by top management.

4.4. Human Resource Information Systems

The human resource management function is concerned with the individuals who constitute the organization. From the standpoint of the organization, the function is responsible for the acquisition and effective use of the individual, the function is concerned with the well being, growth, and development of each worker. To achieve these ends, human resource management departments perform a variety of activities. The major activities of the personnel function in an organization include:

1. recruiting employees
2. evaluating applicants and employees
3. selecting, placing, promoting, terminating, and transferring employees
4. analyzing and designing jobs
5. training and development reports
6. producing required governmental reports
7. managing employee wage and benefit plans
8. planning short and long term staffing needs

To perform these activities, managers rely on a number of operational, tactical, and strategic information systems.

4.4.1. Operational Information Systems

Operation human resource information systems provide the manager with data to support the routine, repetitive personnel decisions that occur regularly. There are several operational level information systems that collect and report personnel data. These include information systems pertaining to the organization's positions and employees, and about governmental regulations.

Payroll Information Systems

The financial information system, through its payroll subsystems, collects and reports data pertaining to human resources that is largely operational in nature. Payroll files often contain a great deal of information about employees – including information about employees pay rates, wage classifications, and seniority – that can be used to managers making human resource decisions.

Position Information Systems

A job is usually defined as a group of like or similar positions. A position, on the other hand, consists of tasks performed by one worker. The purpose of a position information system is to identify each position in the organization, the job category in which the position is classified, and the employee currently assigned to the position. Reference to the position information system allows a personnel manager to identify the details about unfilled positions. Position information systems also allow the personnel manger to identify human resource problems.

Employee Information Systems

The personnel department must maintain information on each of the organization's employee for a variety of reporting purposes. One part of this information system is a personnel file, which usually contains personal and organization related information such as name, address, sex, material status, citizenship, years of service or seniority data, education and training, previous experience, employment history within the organization, salary rate, salary or wage grade, and retirement plans.

Another part of an employee information system is an employee skills inventory. The skills inventory contains information about every employee's work experience, work preference, test scores, interests, and special skills or proficiencies. The skills inventory system provides information for many decisions.

Employee Evaluation Information Systems

Many organizations review the work of employees on a regular basis to make decisions regarding merit pay, pay increases, transfer, or promotion. Typically, a new employee is evaluated at the end of the first four months and other employees are evaluated semiannually. These reviews provided by the employee evaluation information systems are often called performance appraisals. The data for performance appraisals are frequently collected by administering employee appraisals forms to each employee's immediate superior. The forms may also be given to peers, the employees themselves, and even customers or clients.

Applicant Selection and Placement Systems

After job and the employee requirements for those jobs have identified and after a suitable pool of job candidates has been recruited, the candidates must be screened, evaluated, selected and placed in the positions that are open. To ensure compliance with federal, state, and local employment laws, these procedures must be fully documented and carried out in a structured manner. Thus, data pertaining to interviews, examinations, and placement decisions should be collected and kept according to the requirements of the various laws and regulations of acts.

4.4.2. Tactical Human Resource Information Systems

Tactical information systems provide managers with support for decisions that emphasize the allocation of resources. Within the human resource management area, these decisions include recruiting decisions, job analysis and design decisions, training and development decisions, employee compensation plan decision, and labor negotiation decisions.

Job Analysis and Design Information Systems

The information inputs to the job analysis and design information system include data obtained from interviews of supervisors and workers and affirmative action guidelines. Inputs also include information from sources external to the firm, such as labor unions, competitors, and governmental agencies. The outputs of the job analysis information system are job descriptions and job specifications.

Recruiting Information Systems

To develop recruitment plan and monitor its success, a recruiting information system is necessary to collect and process the many different types of information needed to construct the plan. These include a list of unfilled positions; the duties and requirements of these positions; lists of planned employee retirements, transfers or terminations;

information about the skills and preferences of current employees; and summaries of employee appraisals. The source of recruitment, affirmative action plans, information about the success of recent recruitment activities must also be included in the information systems.

Compensation and Benefits Information Systems

Fringe benefit is anything in addition to the basic salary and over time pay to an employee. Information on the benefits and compensation packages will help managers to make informed decisions.

Employee Training and Development Systems

To support the management and training function, it is important to keep information about current and potential instructors; current and potential students; the content and scheduling of courses, seminars, and workshops; and the success of past efforts, as measured by tests, student ratings, and supervisor follow ups. This information system should permit tactical decisions regarding the nature of training and development programs. This system must use the other sources of human resource information systems.

Labor Negotiations Information Systems

Negotiating with the union requires information gathered from many of the human resource information systems already discussed. In addition, negotiators need information from the financial accounting system and from external spruces including competitor wage agreements and appropriate economic data for the industry, employee group, and geographical region.

The personnel team completing the negotiating needs to be able to obtain numerous ad hoc reports that analyze the organization's and union's positions within the framework of the industry and current economic situation. It is also important that the negotiating team be able to receive ad hoc reports on a very timely basis, since additional questions and tactics will occur to the team while the negotiations are underway.

4.4.3. Strategic Human Resource Information Systems

Human resource planning is conducted to ensure that the organization has the right kinds and the right numbers of people at the right places at the right time to achieve its objectives. There are two types of human resource planning that are strategic in nature. These are manpower planning, or staffing planning, and program planning.

Manpower Planning

Organizations that are involved in long term strategic planning, such as those planning to expand into new market areas, or to construct new factories or offices in new locations, or to add new products, will need information about the quantity and quality of the work force that is available to achieve their goal. Manpower planning serves this purpose. This type of planning involves identifying the human resources needed to meet the organizational objectives specified in the strategic plan. This means forecasting the supply and demand of the required work force. These forecasts are estimates of the characteristics, quantity, and pricing of the labor force needed to achieve the long-term plans of the organization.

To forecast human resource needs, information will be required to answer a number of planning questions, including the following:

1. What should the labor force of the organization look like to meet the strategic plan? What kinds of skills, experiences, knowledge, and other qualities should the organization's human resources possess? In other words, what are the job descriptions and specifications required by the strategic plan?
2. What quantities of human resource with the qualities already identified are needed to carry out the strategic plan? In other words, how many positions for each job title are needed for the strategic plan?
3. What are the current human resources of the organization and how do they help the organization's strategic needs for human resources?
4. What other human resources are available to achieve the strategic plan?

Identifying the types and quantities of workers needed for the strategic plan is forecasting the demand for human resources. Identifying the human resources available internally and externally is forecasting the supply of those human resources. Forecasting demand and supply can be done on a macroeconomic level or a microeconomic (more specific to the organization) level.

Program Planning

A second strategic planning function is to develop the policies, procedures, and activities that will achieve the human resource needs spelled out in the manpower plan. Two key activities in program planning are job analysis and design and recruiting. Job analysis and design is essential to describe the work force required by the organization.

Recruiting plans must be developed to acquire the right kinds of workers in the right amounts to fill the jobs described. Where there are insufficient workers with the right skills, training programs may have to develop, or the jobs may have to be redefined to fit the skills of the available work force.

4.5. Managerial Decision Support Systems

A Decision Support System (DSS) is a computer-based information system that supports business or organizational decision-making activities. DSSs serve the management, operations, and planning levels of an organization and help to make decisions, which may be rapidly changing and not easily specified in advance. Decision support systems can be either fully computerized, human or a combination of both.

DSSs include knowledge-based systems. A properly designed DSS is an interactive software-based system intended to help decision makers compile useful information from a combination of raw data, documents, and personal knowledge, or business models to identify and solve problems and make decisions.

Typical information that a decision support application might gather and present includes:

- ✓ inventories of information assets (including legacy and relational data sources, cubes, data warehouses, and data)
- ✓ comparative sales figures between one period and the next
- ✓ Projected revenue figures based on product sales assumptions.
 - Management is decision making
 - The manager is a decision maker
 - Now fast changing, complex environment
 - Factors affecting decision making
 1. Technology/Information/Computers
 2. Structural Complexity/Competition
 3. International Markets/Political Stability
 4. Consumerism/Changes, Fluctuations
- Information
 - ❖ Type of information required is directly related to the level of management and the amount of structure in the decision situation
 - ❖ Levels of managerial decision-making

-
- ❖ Strategic Management
 - ❖ Tactical Management
 - ❖ Operational Management

Monitor the strategic performance of the organization and its overall direction in the political, economic, and competitive business environment

Unstructured Decisions

- ❖ Not possible to specify in advance most of the decision procedures to follow.
- ❖ Decision maker must provide judgment, evaluation and insights to a novel, important and non-routine-type decision.

Require more summarized, ad hoc, unscheduled reports, forecasts, and external intelligence to support their more unstructured planning and policy-making responsibilities

Today it is the competitive, fast business world most organizations start with data processing system that support transaction processing and evolve to management information system which support tactical and strategic level decision making .The opinion is that computer based systems are very valuable for transaction processing and support of operational level decision making.

4.6. Transaction Processing System (TPS)

A transaction is an elementary activity conducted during business operations. Transaction processing systems (TPS) process the company's business transactions and thus support the operations of an enterprise. A TPS records a non-inquiry transaction itself, as well as all of its effects, in the database and produces documents relating to the transaction.

TPS are necessary to conduct business in almost any organization today. TPSs bring data into the organizational databases; these systems are also a foundation on which management oriented information systems rest.

System Charts

Systems charts are well-established tools which are used to describe TPSs. These charts show the sources of input into the system, major processing steps, data storage, and systems outputs.

Transaction Processing Modes

Transaction processing may be accomplished in one of two modes:

1. On-line mode
2. Batch mode

Characteristics of on-line transaction processing:

1. Each transaction is completely processed immediately upon entry.
2. OLAP is the most common mode of used today
3. More costly than batch processing
4. Database is always up to date
5. Require the use of fast secondary storage such as magnetic disks

Characteristics of batch transaction processing:

1. Relies on accumulating transaction data over a period of time and then processing the entire batch at once.
2. Batch processing is usually cyclic: daily, weekly, or monthly run cycle is established depending on the nature of the transactions
3. Cheaper than on-line processing
4. Easier to control than on-line processing
5. Database is constantly out of date
6. Batch processing is now being captured using disk files

Transaction Processing Subsystems in a Firm

Overall transaction processing, also known as data processing, reflects the principal business activities of a firm. The principal transaction processing subsystems in a firm are those supporting:

- | | |
|---------------|------------------------|
| 1. Sales | 7. Accounts payable |
| 2. Production | 8. Billing |
| 3. Inventory | 9. Accounts receivable |
| 4. Purchasing | 10. Payroll |
| 5. Shipping | 11. General ledger |
| 6. Receiving | |

Transaction Processing Activities

The processing of individual transactions, of course, depends to a degree on their nature.

The general elements of transaction processing include:

1. Data capture and validation
2. Transaction - dependent processing steps
3. Database maintenance

Date Capture

Direct data entry is commonly employed through source data automation. Increasingly, transaction processing systems rely on electronic data interchange (EDI). By replacing paper documents with formatted transaction data sent over telecommunications networks, these systems provide for computer-to-computer communication without repeated data entry. Although used internally by some firms, EDI primarily serves the needs of intercompany communication.

Data Validation

Typical validation tests include checking for missing data items, valid codes, and valid values. More extensive validation may entail authorization of the transaction based on the customers' record and available inventory.

Processing Steps Dependent on the Transaction and on Processing Mode

Depending on the nature of the transaction and on whether the system operates in on-line or batch mode, the following processing steps may be performed:

1. **Classification:** The system classifies incoming transactions to select further processing steps.
2. **Sorting Transaction:** records are arranged in order of the value of the data item(s) that uniquely identifies each of them.
3. **Data Retrieval:** The purpose of an inquiry transaction is retrieval of data from the database. Other transactions may involve data retrieval as well.
4. **Calculation:** the calculations required depend on the nature of the transaction.
5. **Summarization:** usually performed to obtain simple reports offered by TPS, this step computes summaries across all or some of the transactions.

Database Maintenance

After transactions other than inquiries, system files or databases must be updated. The data accumulated by TPSs thus serve as a source of detail for management oriented components of information systems.

Outputs Provided by Transaction Processing Systems

The outputs provided by TPSs may be classified as:

1. Transaction documents
2. Query responses
3. Reports

Transaction Documents

Many TPSs produce transaction documents, such as invoices, purchase orders, or payroll checks. These transaction documents produced by TPS may be divided into two classes: action documents and information documents.

1. **Action documents** direct that an action take place. Turnaround documents initiate action and are returned after its completion to the requesting agency. They therefore also serve as input documents for another transaction.

2. **Information documents** confirm that a transaction has taken place or inform about one or several transactions. Transaction documents require manual handling and, in some cases, distribution of multiple copies. The process is costly and may lead to inconsistencies if one of the copies fails to reach its destination.

Query Responses and Reports

TPS offer certain querying and simple reporting capabilities, albeit much less elaborate than those of management reporting systems. Most queries produce a screenful of information. However, reports are also often produced as a result of inquiries.

Unlike management reporting systems, TPSs typically provide a limited range of preplanned reports. The content and format of such reports are programmed into the TPS software and the reports are produced on schedule. The TPS reports are often quite long.

The following report types are produced by TPS

1. **Transaction Logs:** are listings of all transactions processed during a system run and include purchase order manifests or sales registers.
2. **Error (Edit) Reports:** error reports list transactions found to be in error during the processing. They identify the error and sometimes also list the corresponding master file or database records.
3. **Detail Reports:** detail reports are extracts from the database that lists records satisfying particular criteria.
4. **Summary Reports:** typical summary reports produced by TPSs include financial statements.

From Electronic Data Interchange (EDI) to Electronic Commerce

A prominent means of source data automation is electronic data interchange. Electronic data interchange (EDI) is the computer to computer interchange of electronic transaction documents, involving at least two trading partners. With EDI, paper transaction documents, such as purchase orders or invoices are eliminated and replaced with standardized electronic communications. EDI underlines much of electronic commerce

by enabling companies to conclude commercial transactions over telecommunications networks, and the Internet in particular.

EDI components include the following:

1. Transaction standards: the messages are exchanged in a standard form, agreed on by the participating partners.
2. Industry standard for product identification: partners have to agree on the standard way to identify their products.
3. Translation software: translation software converts the incoming EDI messages into a format that can be used by the owner firm's applications.
4. Telecommunications systems: EDI can be carried out via direct telecommunications links between the partners, using a value-added network (VAN) from a third-party supplier, or over the Internet. Among other services, VANs supply electronic mailboxes that can hold messages for the addressee.

Beyond direct savings, EDI has significant potential in competitive and strategic applications of information technology. Among the principal effects are:

1. Compressing the business cycle by speeding up communications
2. Supporting time-based competitive modes, such as the just in time manufacturing strategy that reduces or even removes inventories, and quick response retail strategies
3. Intensified relationships between trading partners. This is due to the cost of switching to another EDI system after the given one is in place and to avoid misunderstandings because of errors, common in the exchanges of paper documents.

Management Reporting Systems

Characteristics of Management Reporting Systems

Management reporting systems are the most elaborate of the management oriented information systems. The main objective of management reporting systems (MRS) is to provide lower and middle management with printed or electronic reports and with inquiry capabilities to help maintain operational and management control of the enterprise.

Characteristics of MRS include:

1. They are usually developed by information systems professionals, rather than by end users, over an extensive period of time, with the use of life cycle oriented development methodologies as opposed to a rapid development by first building a simpler prototype system and then refining it in response to user experience.

-
2. These systems are building for situations in which information requirements are reasonably well known and expected to remain relatively stable.
 3. MRSs do not directly support the decision-making process as a search for alternative solutions to problems and the selection of the solution to be implemented.
 4. MRSs are oriented toward reporting on the past and the present, rather than projecting the future.
 5. MRSs generally have limited analytical capabilities. They are not built around elaborate models, but rather rely on extraction of data from databases according to given criteria, and on summarization of the data.
 6. MRSs largely report on internal company operations rather than spanning the company's boundaries by reporting external information.

Summary

Whether you are the chief operating office of a large international organization or merely the supervisor of a small department at one location in that organization, you will find it necessary to make financial decisions.

Marketing information systems support the major activities of a marketing organization. The information systems collect data that describe marketing operations, process those data, and make marketing information available to marketing managers to help them make effective decisions.

Purpose: manufacturing systems encompasses all the activities necessary to ensure production. These activities may include the evaluation of sites for production; the planning, development, and maintenance of production facilities; and the setting of production goals to meet the requirements of the sales forecast generated by the marketing system.

UNIT 5: KNOWLEDGE MANAGEMENT (KM)

★ Introduction

Knowledge management includes Strategies and processes designed to identify, capture, structure, value, leverage, and share an organization's intellectual assets to enhance its performance and competitiveness. It is based on two critical activities:

- (1) capture and documentation of individual explicit and tacit knowledge, and
- (2) Its dissemination within the organization.

5.1. Introduction to knowledge management

Knowledge management (KM) comprises a range of strategies and practices used in an organization to identify, create, represent, distribute, and enable adoption of insights and experiences. Such insights and experiences comprise knowledge, either embodied in individuals or embedded in organizations as processes or practices. More recently, other fields have started contributing to KM research; these include information and media, computer science, public health, and public policy.

Many large companies and non-profit organizations have resources dedicated to internal KM efforts, often as a part of their business strategy, information technology, or human resource management departments. Several consulting companies also exist that provide strategy and advice regarding KM to these organizations.

Knowledge management efforts typically focus on organizational objectives such as improved performance, competitive advantage, innovation, the sharing of lessons learned, integration and continuous improvement of the organization. KM efforts overlap with organizational learning, and may be distinguished from that by a greater focus on the management of knowledge as a strategic asset and a focus on encouraging the sharing of knowledge. It is seen as an enabler of organizational learning and a more concrete mechanism than the previous abstract research.

Knowledge management is the name of a concept in which an enterprise consciously and comprehensively gathers, organizes, shares, and analyzes its knowledge in terms of resources, documents, and people skills. In early 1998, it was believed that few enterprises actually had a comprehensive knowledge management practice (by any name) in operation. Advances in technology and the way we access and share information has

changed that; many enterprises now have some kind of knowledge management framework in place.

Knowledge management involves data mining and some method of operation to push information to users. A knowledge management plan involves a survey of corporate goals and a close examination of the tools, both traditional and technical that is required for addressing the needs of the company. The challenge of selecting a knowledge management system is to purchase or build software that fits the context of the overall plan and encourages employees to use the system and share information.

The goal of a knowledge management system is to provide managers with the ability to organize and locate relevant content and the expertise required to address specific business tasks and projects. Some knowledge management systems can analyze the relationships between content, people, topics and activity and produce a knowledge map report or knowledge management dashboard.

In an *Information Week* article, Jeff Angus and Jeetu Patel describe a four-process view of knowledge management that we have put into a table:

This major process...	Includes these activities....
Gathering	• Data entry • OCR and scanning • Voice input • Pulling information from various sources • Searching for information to include
Organizing	• Cataloging • Indexing • Filtering • Linking
Refining	• Contextualizing • Collaborating • Compacting • Projecting • Mining
Disseminating	• Flow • Sharing • Alert • Push

KM efforts have a long history, to include on-the-job discussions, formal apprenticeship, discussion forums, corporate libraries, professional training and mentoring programs. More recently, with increased use of computers in the second half of the 20th century, specific adaptations of technologies such as knowledge bases, expert systems, knowledge repositories, group decision support systems, intranets, and computer-supported cooperative work have been introduced to further enhance such efforts.

In 1999, the term personal knowledge management was introduced which refers to the management of knowledge at the individual level.

In terms of the enterprise, early collections of case studies recognized the importance of knowledge management dimensions of strategy, process, and measurement. Key lessons learned included: people and the cultural norms which influence their behaviors are the

5.2. The Theory of Knowledge Management

To manage knowledge in organizations, we have to rely on concepts and constructs that are theoretically sound, which cover the most important areas of knowledge processes, and which are easy to communicate and integrate in the practical action within the organization. In the previous sections we have developed theoretical foundations for knowledge management. As we saw, conceptually robust theories of organizations, knowledge, and meaning processing require rather sophisticated discussions on the nature of intelligence, meaning, organized action, and organizational information processing.

From this theoretical basis, we should now be able to derive theoretical constructs that are directly relevant for practical organizational life. We should also package these theories into a form that can be integrated into management practice. Based on the previous discussion, we should now be able to describe the different types of knowledge in organizations, the ways knowledge is generated, and the ways knowledge integrates with work activities and strategic development of organizational competencies and processes.

In Parts II and III we used a number of theoretical approaches in an attempt to clarify the nature of intelligence and organizations. Loosely speaking, they all can be described as “phenomenological” approaches, in contrast to much of the extant theory that has been based on objectivistic epistemologies, information processing, and cognitivism. Based on those theoretical considerations, I argued that the focal units of organizational knowledge creation can be viewed as communities. Organizations themselves can be conceptualized as almost autopoietic systems whose meaning structure defines what can be information for them at the organizational level of analysis. More fundamentally, however, organizations need to be understood within ecology of social systems. Based on

Luhmann’s analysis of social systems as meaning processing systems, I argued that organizational communities are systems that self-referentially process meaning. Therefore, they can also be called cognitive systems, and the metaphor of organizational intelligence is interesting and appropriate. The coupling between individual cognition and organizational cognition is, however, loose, as humans-in-society and organizations live in 286 phenomenally irreducible worlds. An interesting way to approach these loose couplings is to analyze their time relations.

Organizations and organizational actors can manage their knowledge at the various levels where knowledge exists in the organization. At all these levels we may ask what tools and behaviors increase the possibilities for effective action. Moreover, we may ask whether, for example, we can design and implement organizational structures within the focal organization that increase organizational intelligence.

Vygotsky's observation was that language and conceptual thinking become tools for cognition and simultaneously change it. Language, definitely, is one of those tools that we use to manage knowledge. The meaning structures that underlie language embed major stocks of social and historically developed knowledge. Luhmann, however, pointed out that language, as a media and tool, creates tensions, which, in turn, generate further media. As communication is inter-personal, its success is inherently improbable. To overcome the inherent improbability of communicative success, language emerges with media that release tensions created by the three improbabilities of accessibility, acceptance, and understanding. For example, symbolically generalized meanings and conceptual systems discussed by Vygotsky are, in Luhmann's terms, media that manage tensions in communication.

Organizations, themselves, can also be viewed as tools, and, as social systems, they also embed stocks of knowledge. They are, in the Bergsonian sense, examples of "organized matter," constructed from elements available in the social world. As they are social tools, they can simultaneously be used for multiple purposes by the different members of society, both inside and outside the focal organization. To maintain the organization, these purposes, however, have to be mutually compatible. A single focal actor or motive is not sufficient in explaining the nature of organizations. There may be several actors, and the focus of activity may vary. Indeed, metaphorically, we could view an "organizational tool" as analogous to a boat, which several actors can board for various purposes when they want to navigate toward the same direction. An organization can be emerge through collaborative action, or it can be intentionally designed. In the boat metaphor, the first case would happen when people want to sail across an ocean and join their forces in building a boat, the second case when an individual commissions the construction, and after the boat is ready, sells tickets for the journey. Probably the latter better describes 287 traditional industrial organizations, whereas the previous more closely describes a modern "knowledge-intensive" organization.

The mediated cognition view was based on Bergson's and Vygotsky's analysis of intelligence, language, and mediated thought. Intelligence can be defined as the process that generates meaning structures, which, in turn, underlie effective action. In this terminology, intelligence is a continuous process, and knowledge its accumulated product. Here, as in all biological life, several simultaneous processes operating in different time-scales both produce and reproduce the system. Intelligence recursively defines itself in the process that simultaneously operates within the existing meaning structure and changes it. Metaphorically, we could then say that intelligence is not something that we "have," or something that "is"; instead, it is a process in time that enables "becoming," in true Bergsonian and biological sense. Some parts of the meaning structure provide the "background" for meaning processing, and these relatively "institutionalized" parts of the meaning structure we can call knowledge structures. Other parts of the meaning structure change when the information in the environment changes.

This we could call perception. Perception and knowledge, therefore, are not fundamentally different. Instead, they both define what a meaningful reality is for an intelligent being. There is no fixed privileged position for "institutionalized" meaning structures, except the fact that they are actively reproduced as much of the meaning processing relies on them. Some core concepts and knowledge is central to the reality we operate in, and their reinterpretation requires a paradigm shift that rearranges a large number of meaning relations. As soon as meaning structure changes so that old "institutionalized" meaning structures are not recreated, they disappear and new knowledge emerges. In the pragmatist view, knowledge changes when experience so requires; however, when we compare knowledge with perception, we can say that those meaning relations that underlie knowledge are more "sticky" than those that dynamically become organized in the process of perception.

If we reserve the word "intelligence" for the process of meaning processing, and the word "knowledge" for relatively stable accumulated meaning structures, cognition, in its broadest sense, can be defined as capability for effective choice. Cognition, therefore, also means capability to create information about the environment. What "effectiveness" in each case means, depends on the acting unit, and there are no universal criteria for it. Within the autopoietic framework one could, however, say that to be effective, action has to maintain the system organization, although it can at the same time change its structure.

Within the activity theoretic framework, we could say that effectiveness of activity is measured by the correspondence between the needs of the actor and their fulfillment by the activity.

In contrast to objectivistic theories of knowledge, we would not—and could not—define knowledge as “true justified belief” unless we completely redefine the concept of truth, as for example Polanyi did. This is simply because we know the world in the same way as its facts: through socially constructed and historically developed distinctions. The criterion for “truth” and knowledge is therefore pragmatic and defined only within a community of thought. The experts in the community define what knowledge is for the community, but their role as experts, in turn, is defined by the community. Therefore, knowledge evolves in the same way as the meaning of a concept changes every time it is used in meaning processing. Knowledge is therefore not fixed to any objective reality.

Neither is knowledge subjective or truth purely relativistic. To put it in other words, our knowledge cannot strictly speaking be “false”; instead, it can only make our behavior “stupid” and incompetent. In some cases our incompetence is measured by other social observers, in other cases we just unexpectedly hit our heads into some natural walls. The third theoretical perspective, the developmental view, focused on the ways that knowledge changes and accumulates. In the Vygotskian framework, the three lines of phylogenetic, ontogenetic, and cultural development interact. Learning occurs through joint effort among people who share a culture or praxis. Practical intelligence uses tools that embed knowledge about practice, and intelligence is augmented by cognitive tools. Identities of people are bound to cultures and communities of practice that interacting and communicating people mutually construct. Simultaneously, knowledge also becomes defined in relation to these social formations. What counts as effective action depends on tools and practices available within a community, as well as on sedimented social structures.

In the developmental view, cognition, knowledge, and intelligence are not stable. The development of knowledge structures changes the way intelligence functions. Simultaneously it changes the criteria for effectiveness. Ontogenic change can lead to new effective habits and concepts; and concepts, in turn, can sediment into structural knowledge. World is continuously constructed using language and socialization, and this emerging world is embedded in new practices, tools, and social structures.

5.3. KM Sharing

Knowledge Sharing is an activity through which knowledge (i.e., information, skills, or expertise) is exchanged among people, friends, families, communities (e.g., Wikipedia), or organizations.

Organizations have recognized that knowledge constitutes a valuable intangible asset for creating and sustaining competitive advantages. Knowledge sharing activities are generally supported by knowledge management systems. However, technology constitutes only one of the many factors that affect the sharing of knowledge in organizations, such as organizational culture, trust, and incentives. The sharing of knowledge constitutes a major challenge in the field of knowledge management because some employees tend to resist sharing their knowledge with the rest of the organization.

Although knowledge is commonly treated as an object, Snowden has argued it is more appropriate to teach it as both a flow and a thing. Knowledge can be related to the concept of tacit knowledge, discovered by Ludwik Hirsztfeld which was later further explicated by Nonaka. While the difficulty of sharing knowledge is in transferring knowledge from one entity to another, it may prove profitable for organizations to acknowledge the difficulties of knowledge transfer and its paradoxicality, adopting new knowledge management strategies accordingly.

Explicit knowledge sharing occurs when explicit knowledge is made available to be shared between entities. Explicit knowledge sharing can happen successfully when the following criteria are met:

- **Articulation:** the knowledge provider can describe the information.
- **Awareness:** the recipient must be aware that knowledge is available.
- **Access:** the knowledge recipient can access the knowledge provider.
- **Guidance:** the body of knowledge must be defined and differentiated into different topics or domains so as to avoid information overload, and to provide easy access to appropriate material. Knowledge managers are often considered key figures in the creation of an effective knowledge sharing system.
- **Completeness:** the holistic approach to knowledge sharing in the form of both centrally managed and self-published knowledge.

Tacit knowledge sharing occurs through different types of socialization. Although tacit knowledge is difficult to identify and codify, relevant factors that influence tacit knowledge sharing include:

- Informal networks such as daily interactions between people within a defined environment (work, school, home, etc.). These networks span hierarchies and functions.
- The provision of space where people can engage in unstructured or unmonitored discussions, thereby fostering informal networks.
- Unstructured, less-structured or experimental work practices that encourage creative problem solving, and the development of social networks.

Embedded knowledge sharing occurs when knowledge is shared through clearly delineated products, processes, routines, etc. This knowledge can be shared in different ways, such as:

- Scenario planning and debriefing: providing a structured space to create possible scenarios, followed by a discussion of what happened, and how it could have been different.
- Management training.
- Knowledge transfer: deliberately integrating systems, processes, routines, etc., to combine and share relevant knowledge.

Knowledge constitutes a valuable, intangible asset for creating and sustaining competitive advantages within organizations. Several factors affect knowledge sharing in organizations, such as organizational culture, trust, incentives, and technology. Knowledge sharing activities are commonly supported by knowledge management systems, a form of information technology (IT) that facilitates and organizes information within a company or organization.

Knowledge sharing can sometimes constitute a major challenge in the field of knowledge management. The difficulty of knowledge sharing resides in the transference of knowledge from one entity to another. Some employees tend to resist sharing their knowledge because of the notion that knowledge is property; ownership, therefore, becomes very important. In order to counteract this, individuals must be reassured that they will receive some type of incentive for what they create. However, individuals are most commonly rewarded for what they know, not what they share. Negative consequences, such as isolation and resistance to ideas, occur when knowledge sharing is

impeded. To promote knowledge sharing and remove knowledge sharing obstacles, the organizational culture of an entity should encourage discovery and innovation.

5.4. Technologies to Support KM

Knowledge Management requires technologies to support the new strategies, processes, methods and techniques to better create, disseminate, share and apply the best knowledge, anytime and anyplace, across the team, across teams, across the organization and across several organizations, especially its clients, customers, partners, suppliers and other key stakeholders.

The key technologies are communication and collaboration technologies that are web based for internet and intranet usage, as well as mobile technologies such as PDA's, PC's, telephone and videoconferencing. New technologies are rapidly emerging that act as intelligent agents and assistants to search, summarize, conceptualize and recognize patterns of information and knowledge.

For an effective KM initiative across the organization, there needs to be in place, at least:

- **Knowledge Portal**

There is often confusion between the terms 'information portal' and 'knowledge portal'.

An information portal is often described as a gateway to information to enable the user to have one, more simplified way of navigating towards the desired information.

However a 'knowledge portal' is far more than an information portal because, as well as information navigation and access, it contains within it software technologies to, at least, support the processes of virtual team communication and collaboration and software technologies to support the 9 step process of managing knowledge. Furthermore, it contains intelligent agent software to identify and automatically distribute information and knowledge effectively to knowledge workers based on knowledge profiling.

- **Knowledge Profiles**

Within the knowledge portal, each knowledge worker can update and maintain a personal 'knowledge profile' which identifies his/her specific knowledge needs, areas of interest and frequency of distribution.

- **Collaborative workspaces**

Within the knowledge portal, shared work spaces can be set up for each new team or project. These will become knowledge repositories from which new knowledge will be distilled regularly and systematically and shared across other teams in the organization.

Within the shared and collaborative workspace, at least, the following communication and collaboration functions could be performed:

- | | |
|---|--|
| ✓ Shared vision and mission | ✓ Document libraries |
| ✓ Specific team objectives | ✓ Discussion forums |
| ✓ Knowledge Plan | ✓ Centralized email |
| ✓ Team members roles and responsibilities | ✓ Capturing of new learning's and ideas |
| ✓ Team contract | ✓ Peer reviews, learning reviews, after action reviews |
| ✓ Best Knowledge Bases or Banks | ✓ New knowledge nominations |
| ✓ Expert locator | |
| ✓ Task management | |
| ✓ Shared Calendar management | |
| ✓ Meeting management | |

▪ **Urgent requests**

Within the knowledge portal, it is very useful to have a facility and underlying process to enter any 'Urgent Request' into the portal and receive back any responses from across the organization. Rather than needing to know 'who might know' the request is entered blindly and responses will be made if it is known in the organization and people are willing to support and respond to this activity. This is a very effective way of better leveraging the knowledge across the organization.

▪ **Document Libraries**

The document library is typically the location where all documents are stored. The library should be context relative and allow the ease of control over any document type. Many organizations now employ an Electronic Document and Records Management System (EDRMS) for this requirements but the integration of the EDRMS with all other relevant information and knowledge sources is imperative.

▪ **Knowledge Server and services**

In order to foster knowledge networking across the entire organization and support knowledge processes for creating, retaining, leveraging, reusing, measuring and optimizing the use of the organizational knowledge assets, a centralized knowledge server is required that will:

- manage the communications and collaboration between networks of people
- enable the access, creation and sharing of knowledge between them

The centralized knowledge server will manage the processes and knowledge services that generate and disseminate knowledge assets.

The key components of a generic knowledge server are:

- ✓ a knowledge portal interface designed around a knowledge asset schema (see KM consulting section) as a gateway to user access, security and applications
- ✓ Knowledge banks
- ✓ Advanced search capabilities
- ✓ collaboration services
- ✓ search and discovery services
- ✓ publishing services based on user knowledge needs and knowledge profiling
- ✓ a knowledge map (taxonomy)
- ✓ knowledge repository for information and process management
- ✓ Text summarizing and conceptualizing
- ✓ Intelligent agent ware
- ✓ an Intranet infrastructure for integrated email, file servers, internet/intranet services

Knowledge Bases (Banks)

For each key knowledge area identified, there needs to be a Knowledge Base.

A Knowledge Base contains:

- both structured and unstructured discussion forums
- rich 'knowledge objects' that have been designed for the efficient and effective transfer of knowledge using multimedia, video, audio
- embedded communications theory (eg storytelling)
- KM processes to:
 - critically review knowledge nominations and turn them into improved knowledge
 - automatically find and publish knowledge to users according to users knowledge profiles
- transfer knowledge effectively

5.5. KM Application in Organizations

The turbulent business environment, with global trends and the stringent competition in the world market resulted in the significant changes that companies should introduce in their current business operations. The changes are necessary for survival in the market. The companies should adapt to changes in the external environment and make changes in

the internal environment in terms of innovation, increasing competitiveness, employee training, increasing flexibility, improving processes and business performance. In a “knowledge economy”, the intangible resources are significant resources that add the value. Since the most business processes are viewed through knowledge management, the real challenge, therefore, is developing an adequate system of knowledge management in companies. How much is the importance of knowledge transfer through organizations, it is confirmed by Balanced Scorecard - learning and growth perspective. Its main goals are to raise skill level in performing internal processes, harmonization among employees and companies’ objectives, improving information systems etc. To increase and maintain the value of intellectual capital, companies invest in education and continual training of its personnel, establish a system of grading employees and their performance, reward and promote quality staff. Obviously, there are the difficulties in carrying out modern systems, especially in developing countries, which include the Serbia. However, there is real hope that local companies will soon experience all the benefits of new systems in practice.

Generally speaking, the knowledge management is the process through which organizations generate value from their intellectual capital and knowledge-based assets. Usually, the value is obtained by finding what employees, partners and customers know, and sharing information with employees, departments and even with other companies, in order to find best practices. For companies is the most important to understand 'what they know'. This knowledge is contained in databases, research and development activities, competent staff and quality products that are supplied in the market.

The knowledge management process is an everyday business process within the company, with major role in transfer of relevant information for decision-making across all levels of structure and corporate governance. The ultimate result of well-created process of knowledge management is that every employee in the company fulfills its mission, which reaches the corporate objectives and strategies, and identifies the most valuable knowledge from the “sea of information”. It is not an easy task because it involves the management structure at the highest levels of management. They are responsible for the processes of finding, selecting, organizing and presentation of information in a manner that promotes understanding of employees in a particular area of interests. Knowledge management is important as a managerial tool, which promotes the

creation of new knowledge and it's sharing through the corporate values. Managers need to have a greater sense of invisible and intangible assets of people, featured in the minds and experiences of employees. Without these assets, companies are unequipped with vision and ability to predict the future. The using of knowledge management process increases the effectiveness of decision-making processes, as well as the level of operational efficiency, flexibility, commitment and involvement of employees.

Knowledge management is not only associated with managing knowledge as a resource, but also to manage business processes that take place using that resource. It should involve the analysis of existing knowledge as a resource, as well as defining the objectives regarding the generation, protection and application of new knowledge, then transfer, exchange and dissemination of knowledge, effective use of knowledge and performance measurement. From external sources, the knowledge is generated by purchasing technology and software, hiring experts, using consultants and strategic partnerships. Internal creation of knowledge is a process of individual learning in a group of individuals as well as the process of "organizational learning". In both cases, the key role is on the company's (organizational) units of research and development and on the units in charge of education and training of employees.

The knowledge management process aims to support innovation and encourage the free flow of ideas through the company. It helps increasing revenues (because the products and services are delivered to market faster) and reducing costs (because it eliminates redundant and unnecessary business processes). This process increases the time that employees spend in the company, because their knowledge and efforts are valued by the system of rewards. Ultimately, the knowledge management process increases the value of the company and its competitiveness as a whole, because it increases the efficiency and effectiveness, the relationship of all resources and innovation. In the knowledge management process, there are four main abilities that refer to skills acquisition, assimilation and transformation of knowledge, and ability to use and exploit knowledge.

The creation (or acquisition) of knowledge involves market research, product, services and process research and development and strategic partnerships. The purpose of the knowledge assimilation is using the knowledge in the right way, which reduce gap between the knowledge that is needed and the knowledge that is currently available. The

transformation of knowledge includes the ability to round up a set of knowledge that was previously segmented and disjointed, or to combine elements of knowledge in new ways. This process aims to create an entirely new knowledge, providing fresh ideas for further application of knowledge and new solutions to meet market needs. The connection between knowledge assimilation and transformation is particularly important because the greater assimilation of knowledge, knowledge transformation can be done more effectively.

The ability to use knowledge includes the company's ability to incorporate knowledge into their business processes, and it is reflected in the development of ideas on how, when and where knowledge can be used to meet market needs.

Successful knowledge management (KM) enables organizations to make increasingly more intelligent decisions. Modern competition places the market battles in terms of manufacturer cost and customer value. Knowledge is a significant business asset that prevents organizations from spending time reinventing solutions, and maintains organizational focus on providing a high quality output. This is accomplished through the storing, sharing, and using of knowledge, with the intent of improving the knowledge throughout the management cycle.

Following assessment, the knowledge can be shared. It must be contextualized so that the shared knowledge can again be enhanced and measured. Contextualization links the learner, the expert knower, and the knowledge outcomes. Contextualization provides a suggested path for the application of the knowledge. Where the knowledge came from - the author, the subject of work, the time and location - and where the knowledge can be sent - an organizational memo to lessen sick days via exercise, or the description of a core capability to a client - makes contextualization important for the next time the knowledge is to be shared.

In the final transport of knowledge, it is acquired and put to use by learners. After this application occurs, the learners may have new insights to increase the value of the knowledge. Some of the information can be tacit. Learners could recall that a machine

worked better after further adjustments, or perhaps they combined the knowledge of several sales strategies with even more success than either strategy on its own. The information used in updating knowledge can also be explicit. For example, a presentation which suggests methods for improvement or a customer review of how the applied knowledge affected them is already codified. Each of these decision points- assessment, contextualization, and updating - use new information to validate and enhance, or divest the existing knowledge, and create a more intelligent, more competitive organization.

The importance of a feedback loop has been mentioned with its emphasis in maintaining the value and validity of a knowledge object. The information needed to assess the knowledge is not the only important characteristic. What occurs when an expert is wrong? How can it be verified that an expert does not distort or withhold knowledge in order to protect what they feel should not be shared? Knowledge is limited by how many experts understand it, and how willing they are to participate in sharing it. Evans (2010) discusses the reality of conflict between an individual's privacy, and an organization's knowledge management practices. Besides the struggles of fostering personal participation in knowledge collection, Evans also discusses potential legal issues with copyrights, patents, and tools like social network analysis which could violate privacy laws. Although Dalkir does not delve into the legal issues, she expresses similar views related to the failing of a KM initiative. A knowledge worker will share no more than they are willing or motivated to contribute. Dalkir (2005) argues the knowledge worker should be persuaded to share in order to gain newer perspectives, and thus gain the maximum experience from the job. Organizations that operate in a transparent manner should be able to worry less about KM legalities and more about how to effectively implement the culture and technological infrastructure that KM needs to be successful.

UNIT 6: ETHICAL AND SECURITY ISSUES

★ Introduction

Dilemmas regarding the life of information are becoming increasingly important in a society that is defined as "the information society". Information transmission and literacy are essential concerns in establishing an ethical foundation that promotes fair, equitable, and responsible practices. Information ethics broadly examines issues related to ownership, access, privacy, security, and community.

Information technology affects common issues such as copyright protection, intellectual freedom, accountability, privacy, and security. Many of these issues are difficult or impossible to resolve due to fundamental tensions between Western moral philosophies (based on rules, democracy, individual rights, and personal freedoms) and the traditional Eastern cultures (based on relationships, hierarchy, collective responsibilities, and social harmony). The multi-faceted dispute between Google and the government of the People's Republic of China reflects some of these fundamental tensions.

6.1. Ethical and Security issues

The use of IT presents major security challenges, poses serious ethical questions, and affects society in significant ways. IT raises ethical issues in the areas of..

- Crime
- Privacy
- Individuality
- Employment
- Health
- Working conditions

But, IT has had beneficial results as well. So as managers, it is our responsibility to minimize the detrimental effects and optimize the beneficial effects.

Basic categories of ethical issues are: -

- Employee privacy
- Security of company records
- Workplace safety

Theories of corporate social responsibility

1. Stockholder theory: - Managers are agents of the stockholders. Their only ethical responsibility is to increase profit without violating the law or engaging in fraud.

-
2. Social Contract Theory: - Companies have ethical responsibilities to all members of society, which allow corporations to exist based on a social contract.
 - First condition – companies must enhance economic satisfaction of consumers and employees
 - Second condition – avoid fraudulent practices, show respect for employees as human beings, and avoid practices that systematically worsen the position of any group in society
 3. Stakeholder theory: - Managers have an ethical responsibility to manage a firm for the benefit of all its stakeholders.
 - Stockholders
 - Suppliers
 - Employees
 - Local community
 - Customers

Sometimes stakeholders are considered to include

 - Competitors
 - Government agencies and special interest groups
 - Future generations

Technology Ethics

There are four Principles concerning technology ethics:

1. Proportionality
 - Good must outweigh any harm or risk
 - Must be no alternative that achieves the same or comparable benefits with less harm or risk
2. Informed consent: - Those affected should understand and accept the risks.
3. Justice: - Benefits and burdens should be distributed fairly
4. Minimized Risk: - Even if judged acceptable by the other three guidelines, the technology must be implemented so as to avoid all unnecessary risk

Ethical Guidelines

- Responsible end users
 - Act with integrity
 - Increase their professional competence
 - Set high standards of personal performance
 - Accept responsibility for their work
 - Advance the health, privacy, and general welfare of the public
- Association of Information Technology Professionals (AITP) definition includes

-
- The unauthorized use, access, modification, and destruction of hardware, software, data, or network resources
 - Unauthorized release of information
 - Unauthorized copying of software
 - Denying an end user his/her own hardware, software, data, or network resources
 - Using or conspiring to use computer or network resources to illegally obtain info or tangible property
 - Hacking
 - The obsessive use of computers, or the unauthorized access and use of networked computer systems
 - Cyber Theft
 - Involves unauthorized network entry and the fraudulent alteration of computer databases
 - Unauthorized use at work
 - Also called time and resource theft
 - May range from doing private consulting or personal finances, to playing video games, to unauthorized use of the Internet on company networks
 - Software Piracy
 - Unauthorized copying of software
 - Software is intellectual property protected by copyright law and user licensing agreements
 - Piracy of intellectual property
 - Other forms of intellectual property covered by copyright laws
 - Music
 - Videos
 - Images
 - Articles
 - Books
 - Other written works
 - Computer viruses and worms
 - Virus

A program that cannot work without being inserted into another program
 - Worm

A distinct program that can run unaided
 - IT makes it technically and economically feasible to collect, store, integrate, interchange, and retrieve data and information quickly and easily.
-

-
- Benefit – increases efficiency and effectiveness
 - But, may also have a negative effect on individual's right to privacy
 - Examples of important privacy issues
 - Accessing private e-mail and computer records & sharing information about individuals gained from their visits to websites and newsgroups
 - Always knowing where a person is via mobile and paging services
 - Examples of important privacy issues (continued)
 - Using customer information obtained from many sources to market additional business services
 - Collecting personal information to build individual customer profiles

The Domain of Ethics

Information technology is a powerful tool that can be used to further organizational goals, pursue national interest, or support environmentally sustainable development. The same technology has also made it easier to engage in ethical or unethical business practices electronically anywhere in the world. The way the technology is deployed in organizations depends on our decisions as managers, computing professionals, and users of information systems. All of us therefore, should make these decisions guided not only by the organizational and technological aspects of information systems, but also in consideration of their effects on individuals.

Ethics and Codes of Ethics

Instructors can begin this chapter by asking students the questions posed in this section of the chapter?

1. How do ethical issues relate to legal ones?
2. Is an ethical breach punishable by law?
3. If there is no law that penalizes a certain conduct, does it mean that you are always right behaving in that way?

Ethics is a study of the principles of right and wrong that ought to guide human conduct.

Human behavior and decision making fall into three domains of:

1. Legal issues
2. Ethical issues
3. Discretionary domain

As we develop and use information systems to solve organizational problems or to respond to opportunities, we need to make sure that our solutions is proper with respect to each of these domains.

Legal Domain - governs a variety of relatively well-described behaviors', specified by law and enforceable in the courts of a given country or within a local jurisdiction.

Ethical Issues - is governed by the general norms of behaviour and by specific codes of ethics. Ethical considerations go beyond legal liability.

Sunshine principle: To see whether your decision making in a given case involves an ethical issue, apply the A sunshine principle@: What if I read about my decisions and subsequent actions in tomorrow's paper?

Discretionary Domain - only if the action is both legal and ethical falls into this domain, where we act properly entirely according to our preferences. In this domain, we can apply such criteria as organizational desirability or cost-benefit analysis.

Knowledge of ethics as it applies to the issues arising from the development and use of information systems helps us make decisions in our professional life. Professional knowledge is generally assumed to confer a special responsibility within its domain. This is why the professions have evolved codes of ethics, that is, sets of principles intended to guide the conduct of the members of the profession.

The principal codes of ethics for information systems professionals are the:

1. ACM Code of Ethics and Professional Conduct
2. The Code of Ethics and Standards of Conduct adopted by DPMA.

These codes of ethics provide guidelines for ethical conduct in the development and use of information technology. End users and IS professionals would live up to their ethical responsibilities by voluntarily following such guidelines. For example, you can be a responsible end user by:

1. Acting with integrity
2. Increasing your professional competence
3. Setting high standards of personal performance
4. Accepting responsibility for your work
5. Advancing the health, privacy, and general welfare of the public

Ethical Theories

Ethical theories give us the foundation from which we can determine what course of action to take when an ethical issue is involved. At the source of ethics lies the idea of reciprocity. There are two fundamental approaches to ethical reasoning:

1. Consequentiality theories

- tells us to choose the action with the best possible consequences. Thus, the utilitarian theory that represents this approach holds that our chosen action should produce the greatest overall good for the greatest number of people affected by our decision. This approach is often difficult to apply, since it is not easy to decide what the Agood@ is and how to measure and compare the resulting Goods@.

2. Obligation (deontological) theories

- argues that it is our duty to do what is right. Your actions should be such that they could serve as a model of behavior for others - and, in particular, you should act as you would want others to act toward you. Our fundamental duty is to treat others with respect, and thus not to treat them solely as a means to our own purposes.

Treating others with respect, means not violating their rights, the principal individual rights are:

1. The right to life and safety
2. The right of free consent
3. The right to privacy
4. The right to private property
5. The right of free speech
6. The right of fair treatment
7. The right to due process

Scenarios in the Ethical Domain

This section describes three small mini cases. Give the students a few minutes to read these issues. Spend a few minutes in open discussion with the students on how they would react to the given situation.

The welfare of individuals and their specific rights need to be safeguarded in the environment of an information society. The principal ethical issues of concern with regard to information systems have been identified as the issues of:

-
1. Privacy
 2. Accuracy
 3. Property
 4. Access

Tracing an ethical issue to its source and the understanding of which individual rights could be violated helps understand the issue.

Privacy

Privacy is the right of individuals to retain certain information about them without disclosure and to have any information collected about them with their consent protected against unauthorized access.

Invasion of privacy is a potent threat in an information society. Individuals can be deprived of opportunities to form desired professional and personal relationships, or can even be politically neutralized through surveillance and gathering of data from the myriad databases that provide information about them.

The Privacy Act serves as a guideline for a number of ethics codes adopted by various organizations. The Act specifies the limitations on the data records that can be kept about individuals. The following are the principal privacy safeguards specified:

1. No secret records should be maintained about individuals
2. No use can be made of the records for other than the original purposes without the individuals consent.
3. The individual has the right of inspection and correction of records pertaining to him or her.
4. The collecting agency is responsible for the integrity of the record-keeping system

The power of information technology to store and retrieve information can have a negative effect on the right to privacy of every individual. Computers and related technologies enable the creation of massive databases containing minute details of our lives which can be assembled at a reasonable cost and can be made accessible anywhere and at any time over telecommunications network throughout the world.

Two database phenomena create specific dangers.

Database matching - makes it possible to merge separate facts collected about an individual in several databases. If minute facts about a person are put together in this

fashion in a context unrelated to the purpose of the data collection and without the individual's consent or ability to rectify inaccuracies, serious damage to the rights of the individual may result.

Statistical databases - are databases that contain large numbers of personal records, but are intended to supply only statistical information. A snooper, however, may deduce personal information by constructing and asking a series of statistical queries that would gradually narrow the field down to a specific individual.

Legislation and enforcement in the area of privacy in the United States are behind those in a number of other countries. The countries of the European Union offer particularly extensive legal safeguards of privacy. In the environment of business globalization, this creates difficulties in the area of trans border data flow, or transfer of data across national boundaries. Countries with more stringent measures for privacy protection object to a transfer of personal data into the states where this protection is more lax. The United Nations has stated the minimum privacy guarantees recommended for incorporation into national legislation.

Privacy protection relies on the technical security measures and other controls that limit access to databases and other information stored in computer memories or transmitted over the telecommunication networks.

Accuracy

Pervasive use of information in our societal affairs means that we have become more vulnerable to misinformation. Accurate information is error-free, complete, and relevant to the decisions that are to be based on it.

Professional integrity is one of the guarantors of information accuracy. An ethical approach to information accuracy calls for the following:

1. A professional should not misrepresent his or her qualifications to perform a task.
2. A professional should indicate to his or her employer the consequences to be expected if his or her judgment is overruled
3. System safeguards, such as control audits are necessary to maintain information accuracy. Regular audits of data quality should be performed and acted upon.

-
4. Individuals should be given an opportunity to correct inaccurate information held about them in databases.
 5. Contents of databases containing data about individuals should be reviewed at frequent intervals, with obsolete data discarded.

Property

The right to property is largely secured in the legal domain. However, intangibility of information is at the source of dilemmas which take clarity away from the laws, moving many problems into the ethical domain. At issue primarily are the rights to intellectual property: the intangible property that results from an individual's or a corporation's creative activity.

Intellectual property is protected in the United States by three mechanisms:

1. Copyright

- A method of protecting intellectual property that protects the form of expression (for example, a given program) rather than the idea itself (for example, an algorithm).

2. Patent

- Method of protecting intellectual property that protects a non-obvious discovery falling within the subject matter of the Patent Act.

3. Trade secret

- Intellectual property protected by a license or a non-disclosure agreement

Computer programs are valuable property and thus are the subject of theft from computer systems. Unauthorized copying of software (software piracy) is a major form of software theft because software is intellectual property which is protected by copyright law and user licensing agreements.

Access

It is the hallmark of an information society that most of its workforce is employed in the handling of information and most of the goods and services available for consumption are information-related. Three necessities for access to the benefits of an information society include:

1. The intellectual skills to deal with information
2. Access to information technology
3. Access to information

One should strive to broaden the access of individuals to the benefits of information society. This implies broadening access to skills needed to deal with information by further enabling literacy, access to information technology, and the appropriate access to information itself.

Intensive work is being done on developing assistive technologies - specialized technologies that enhance access of the handicapped to the information technology and, in many cases, to the world at large.

Making Ethical Decisions

This section continues with the three small mini cases from section 17.2. Give the students a few minutes to read these issues. Spend a few minutes in open discussion with the students on how they can follow the decision-making sequences outlined in the text.

1. Examine the issues to see whether they fall in the ethical domain. Seek appropriate professional guidance if the issues seem to be in the legal domain. Apply the Asunshine test@ to see whether the issue is ethical or discretionary: would your decision withstand a public disclosure?
2. If you believe that ethical issues are involved, would a course of action you are considering violate individual rights? Would the action violate a professional code of ethics?
3. Choose the course of action that would not result in an ethical breach.

Impacts of Information Technology on the Workplace

Due to the pervasive use of information technology and its dual potential to be used for good or bad, we need to consider the specific issues that arise when people work with information systems.

Positive and Negative Potential of Information Technology

The ACM Code of Ethics and Professional Conduct commits computer professionals to a design and build information systems that enhance the quality of working life.@

It has been established that people experience job satisfaction when:

1. They have a sense that their work is meaningful
2. They feel a sense of responsibility for the results of their work and have a sense of autonomy and control
3. They receive feedback about their accomplishments

Socio-technical design of information systems is performed in recognition of these crucial factors of employee motivation. Information systems with very similar functionalities can have positive or negative consequences in the workplace. The same information technology can have different impacts, depending on the way it is used in an organization.

Some of the negative effects of information technology include:

1. Use of computers has displaced workers in middle management (whose primary purpose was to gather and transfer information) and in clerical jobs.
2. Some categories of work have virtually disappeared which has created unemployment for a number of workers
3. May create a permanent underclass who will not be able to compete in the job market
4. Computer crime is a growing threat (money theft, service theft, software theft, data alteration or theft, computer viruses, malicious access, crime on the internet).
5. Health issues
6. Societal issues (privacy, accuracy, property, and access)

Some of the positive effects of information technology include:

1. The ability to work from remote locations.
2. Access to individuals with disabilities
3. Medical diagnosis
4. Computer-assisted instruction (learning aids)
5. Environmental quality control
6. Law enforcement

Computer-Based Work Monitoring

Computer-based work monitoring and, related to it, telephone call accounting and service monitoring is practiced to ensure the quality of customer service and to provide objective evaluation of employee performance. The aggregate information gained from these information systems is necessary for management as a means of planning and control. However, when used improperly, these systems can not only raise ethical concerns but actually be counterproductive.

Invasive use may result in increased employee stress and a sense of lack of autonomy. When the gather information is inappropriately applied to individual employees, such

usage raises questions about privacy and quality of working life. It should be a primary goal of managers and computing professionals to lower the invasiveness of monitoring.

Emerging Technologies: Opportunities and Threats in the Workplace. The emerging new technologies keep offering opportunities to improve the effectiveness and efficiency of people's work - and present new threats to their rights. Health issues - the use of technology in the workplace raises a variety of health issues. Heavy use of computers is reportedly causing health problems such as:

1. Job stress
2. Damaged arm and neck muscles
3. Eye strain
4. Radiation exposure
5. Death by computer-caused accidents

Ergonomics - solutions to some health problems are based on the science of ergonomics, sometimes called human factors engineering. The goal of ergonomics is to design health work environments that are safe, comfortable, and pleasant for people to work in, thus increasing employee morale and productivity. Ergonomics stresses the healthy design of the workplace, workstations, computers and other machines, and even software packages. Other health issues may require ergonomic solutions emphasizing job design, rather than workplace design.

Whistle blowing - According to the full text of the ACM Code of Ethics and Professional Conduct, a computing professional has an obligation to report any signs of systems dangers that might result in serious personal or social damages. If one's superiors do not act on the warning, the professional may find it his or her ethical obligation to report the violations outside of the organization. However, an extremely careful assessment of all the relevant aspects of risk and responsibility must precede such an act. Otherwise, the reporting itself may be harmful. Ethical behavior of employees is highly dependent on the corporate values and norms - on the corporate culture as a whole. Open debate of ethical issues in the workplace and continuing self-analysis help keep ethical issues in focus. Many corporations have codes of ethics and enforce them as part of a general posture of social responsibility.

